Safe Entry and Cleaning of Petroleum Storage Tanks

Planning and Managing Tank Entry from Decommissioning Through Recommissioning

API STANDARD 2015
FIFTH EDITION, MAY 1994

STEP
Strategies for Today's Environmental Partnership

American Petroleum Institute
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STEP

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- To counsel customers, transporters and others in the safe use, transportation and disposal of our raw materials, products and waste materials.
- To economically develop and produce natural resources and to conserve those resources by using energy efficiently.
- To extend knowledge by conducting or supporting research on the safety, health and environmental effects of our raw materials, products, processes and waste materials.
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- To work with others to resolve problems created by handling and disposal of hazardous substances from our operations.
- To participate with government and others in creating responsible laws, regulations and standards to safeguard the community, workplace and environment.
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Health and Environmental Affairs Department

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FOREWORD

This standard was prepared under the auspices of the API Committee on Refinery Equipment in consultation with the Safety and Fire Protection Subcommittee. It is intended for use by API member companies and others to develop safety practices for planning, managing, and conducting work in atmospheric and low pressure storage tanks.

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Safe Entry and Cleaning of Petroleum Storage Tanks

SECTION 1—GENERAL

1.1 Scope

This standard provides safety practices for preparing, emptying, isolating, ventilating, atmospheric testing, cleaning, entry, hot work, and recommissioning activities in, on, and around atmospheric and low-pressure (up to and including 15 psig) aboveground storage tanks that have contained flammable, combustible, or toxic materials. This standard directs the user from decommissioning (removal from service) through recommissioning (return to service). This standard applies to stationary tanks used in all sectors of the petroleum and petrochemical industry, including crude oil and gas production, refineries, petrochemical plants, and terminals.

Appendixes which provide additional guidance on specific aspects of tank preparation, emptying, isolation, ventilation, atmospheric testing, cleaning, entry, hot work, and recommissioning activities are included as follows:

Appendix A—Description of Petroleum Storage Tank Hazards
Appendix B—Vapor Freeing the Tank
Appendix C—Control of Ignition Sources
Appendix D—Tank Cleaning Precautions and Procedures for Specific Tank Designs and Products
Appendix E—Planning Checklist
Appendix F—Oxygen and Flammable Vapor Analyzers

All appendixes are offered as guidance and do not form a part of this standard.

This standard does not apply to the following types of tanks or vessels:

a. Pressure vessels exceeding 15 psig.
b. Cryogenic or refrigerated tanks.
c. Vessels maintained under a vacuum.
d. Process vessels.
e. Underground storage tanks.

1.2 Regulatory Requirements

This standard is intended to be consistent with Title 29 of the U.S. Code of Federal Regulations (Occupational Safety and Health Administration). In addition to this standard, users shall refer to applicable federal, state, and local regulations pertinent to specific circumstances, including regulations of governments other than those of the United States when appropriate. If any provision of this standard conflicts with statutory or regulatory requirements, the statute and regulations shall govern. This standard does not function as a substitute for review of applicable regulations or company guidelines in their entirety.

1.3 Definitions

1.3.1 *Attendant* means an individual stationed outside one or more permit spaces who monitors the authorized entrants and who performs all attendants duties assigned in the employer's confined space program (see OSHA 29 CFR 1910.146; also see, OSHA 29 CFR 1910.134 concerning the use of standby personnel).

1.3.2 *Blanking or blinding* means the absolute closure of a pipe, line, or duct by the fastening of a solid plate (such as a spectacle blind or skillet blind) that completely covers the bore and that is capable of withstanding the maximum pressure of the pipe, line, or duct with no leakage beyond the plate (see OSHA 29 CFR 1910.146).

1.3.3 *A Class I, Division 1, Group D* location is a location to which any one of the following applies:

a. Ignitible concentrations of flammable gases or vapors can exist under normal operating conditions.
b. Ignitible concentrations of flammable gases or vapors may exist frequently because of repair or maintenance operations or because of leakage.
c. Breakdown or faulty operation of equipment or processes that might release ignitible concentrations of flammable gases or vapors, and might also cause simultaneous failure of electrical equipment.

Group D refers to the level of protection required for products ranging from crude oil, to gasoline, to liquified petroleum gas. Some petrochemical products will require a greater level of protection such as Group A, B, or C or greater (see NFPA 70).

1.3.4 *Confined space* means a space that (1) is large enough and so configured that an employee can bodily enter and perform assigned work (2) has limited or restricted means for entry or exit (for example, tanks, vessels, silos, storage bins, hoppers, vaults, and pits are spaces that may have limited means of entry) and (3) is not designed for continuous employee occupancy (see OSHA 29 CFR 1910.146).

1.3.5 *Double block and bleed* means the closure of a line, duct, or pipe by closing and locking or tagging two in-line valves and by opening and locking or tagging a drain or vent valve in the line between the two closed valves (see OSHA 29 CFR 1910.146).

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1.3.6 An *employer* is an owner, operator, or contractor whose respective employees are performing a task or activity.

1.3.7 *Engulfment* means the surrounding and effective capture of a person by a liquid or finely divided (flowable) solid substance that can be aspirated to cause death by filling or plugging the respiratory system or that can exert enough force on the body to cause death by strangulation, constriction, or crushing (see OSHA 29 CFR 1910.146).

1.3.8 *Entry* means the action by which a person passes through an opening into a permit-required confined space. Entry includes ensuing work activities in that space and is considered to have occurred as soon as a part of the entrant's body breaks the plane of an opening into the space (see OSHA 29 CFR 1910.146).

1.3.9 *External floating-roof tanks* are cylindrical vessels that have a roof that floats on the surface of the liquid stock. In addition to a cylindrical shell, the basic components include (a) a floating roof, (b) an annular rim seal attached to the perimeter of the floating roof, and (c) roof fittings that penetrate the floating roof and serve operational functions.

1.3.10 The *flammable range* is the range of concentrations of flammable vapor in air, between the lower flammable limit and the upper flammable limit that will propagate flame if ignited (see NFPA 325M).

1.3.11 *Internal floating-roof tanks* are cylindrical vessels that have both a fixed roof over the top of the tank and a floating deck that rests on the liquid stock surface. There are two basic types of internal floating-roof tanks: (1) tanks in which the fixed roof is supported by vertical columns within the tank, which are typical of fixed-roof tanks fitted with an internal floating deck; and (2) tanks with a self-supporting fixed roof, with no internal support columns, which are typical of external floating-roof tanks covered with a fixed roof. Tanks initially constructed with both a fixed roof and a floating deck may be of either type.

1.3.12 *Immediately dangerous to life or health (IDLH)* means any condition that poses an immediate or delayed threat to life or that would cause irreversible adverse health effects or that would interfere with an individual's ability to escape unaided from a permit space.

Note: Some materials—hydrogen fluoride gas and cadmium vapor, for example—may produce immediate transient effects that, even if severe, may pass without medical attention, but are followed by sudden, possibly fatal collapse 12–72 hours after exposure. The victim "feels normal" from recovery from transient effects until collapse. Such materials in hazardous quantities are considered to be "immediately" dangerous to life or health.

Note: Certain irritation effects may affect the ability to escape (see OSHA 29 CFR 1910.146).

1.3.13 *Isolation* means the process by which a permit space is removed from service and completely protected against the release of energy or material into the space by such means as the following: blanking or blinding; misaligning or removing sections of lines, pipes, or ducts; a double block and bleed system; lockout or tagout of all sources of energy; or blocking and disconnecting all mechanical linkages (see OSHA 29 CFR 1910.146).

1.3.14 A *lead-hazard-free tank* is either a tank that was never used to store leaded gasoline or a leaded-gasoline storage tank that has been cleaned according to D.4, tested, and found to have an internal atmosphere below the current OSHA permissible exposure limit currently 0.075 milligrams of organic lead per cubic meter (2 micrograms per cubic foot) (see OSHA 29 CFR 1910.1000 and following; see also 58 Federal Register 35338, June 30, 1993).

1.3.15 The *lower flammable limit (LFL)* is the minimum concentration of a flammable vapor in air which will ignite if an ignition source is applied (see NFPA 325M).

1.3.16 A *material safety data sheet (MSDS)* means written or printed material concerning a hazardous chemical which is prepared in accordance with OSHA 29 CFR 1910.1200. MSDSs provide physical properties, safety, and health data.

1.3.17 *Oxygen deficient atmosphere* means an atmosphere containing less than 19.5 percent oxygen by volume (see OSHA 29 CFR 1910.146).

1.3.18 *Oxygen enriched atmosphere* means an atmosphere containing more than 23.5 percent oxygen by volume (see OSHA 29 CFR 1910.146).

1.3.19 *Permissible exposure limit (PEL)* is the OSHA-designated limit of any airborne contaminant to which an employee may be exposed. The PEL may be expressed as an 8-hour time-weighted average, a ceiling value, a short-term exposure limit, or a skin designation (see OSHA 29 CFR 1910.1000 Subpart Z).

1.3.20 *Pyrophoric* designates a material that may spontaneously ignite upon exposure to air.

1.3.21 A *qualified person* is a person or persons designated by an employer as having the necessary training and competence to perform and/or supervise tasks or activities in accordance with the employer's policy, procedures, and programs (see 2.2).

1.3.22 *Shall* is used to designate information presented in this standard which is mandatory.

1.3.23 *Should* is used to designate information in this standard which is a recommendation for consideration.

1.3.24 *Sour* refers to the presence of hydrogen sulfide in petroleum products (for example, sour crude oil and sour crude oil fractions).

1.3.25 *Threshold limit value (TLV)* is the maximum airborne concentration of substance to which, it is believed, nearly all workers may be repeatedly exposed day after day without adverse effect as determined by the responsible committees of the American Conference of Governmental Industrial Hygienists. (See ACGIH booklet *Threshold Limit*

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Values for Chemical Substances and Physical Agents and Biological Exposure Indices and Documentation.)

1.3.26 The *upper flammable limit (UFL)* is the maximum concentration of a flammable vapor in air which will ignite if an ignition source is applied (see NFPA 325M).

1.3.27 *Vapor freeing* means ventilation or purging to remove flammable and/or toxic materials from the tank atmosphere.

1.3.28 A *work permit* is the written system for authorizing and controlling work performed in a facility. The work permit describes the work to be done and serves as a safety and health checklist for each job. Work permits include but are not limited to entry permits, cold work permits, and hot work permits.

1.4 Referenced Publications

The following standards, codes, and publications are useful references related to safe entry and cleaning of petroleum storage tanks. The most recent edition should be consulted, as appropriate.

ACGIH[1]

Threshold Limit Values for Chemical Substances and Physical Agents and Biological Exposure Indices and Documentation (published annually)

ANSI[2]

Z49.1 *Safety in Welding and Cutting*
Z88.2 *American National Standard for Respiratory Protection*

API

Bul E2 *Bulletin on Management of Naturally Occurring Radioactive Materials (NORM) in Oil and Gas Production*
Std 653 *Tank Inspection, Repair, Alteration and Reconstruction*
RP 2003 *Protection Against Ignitions Arising Out of Static, Lightning and Stray Currents*
Publ 2009 *Safe Welding and Cutting Practices in Refineries, Gas Plants and Petrochemical Plants*
Publ 2026 *Safe Descent Onto Roofs of Tanks in Petroleum Service*
Publ 2027 *Ignition Hazards Involved in Abrasive Blasting of Atmospheric Storage Tanks in Hydrocarbon Service*
Publ 2202 *Dismantling and Disposing of Steel From Aboveground Leaded Gasoline Storage Tanks*
Publ 2207 *Preparing Tank Bottoms for Hot Work*
Publ 2219 *Safe Operation of Vacuum Trucks in Petroleum Service*
RP 2220 *Improving Owner and Contractor Safety Performance*

CGA[3]

G-7.1 *Commodity Specification for Breathing Air*

DOT[4]

49 *Code of Federal Regulations* (49 CFR)
Parts 171 – 180 Subchapter C, Hazardous Materials Regulations

EPA[5]

40 *Code of Federal Regulations* (40 CFR)
260 – 271

ICS, IAPH, OCIMF[6]

International Safety Guide for Oil Tankers and Terminals

NFPA[7]

Fire Protection Handbook
51B *Cutting and Welding Processes*
70 *National Electrical Code*
77 *Static Electricity*
325M *Fire Hazard Properties of Flammable Liquids, Gases and Volatile Solids*

NIOSH[8]

2533 *Manual of Analytical Methods*

OSHA[9]

29 *Code of Federal Regulations* (OSHA 29 CFR), including:

1910.38 Employee Emergency Plans and Fire Prevention Plans
1910.95 Noise
1910.106 Flammable and Combustible Liquids
1910.120 Hazardous Waste Operations and Emergency Response
1910.132-6 Subpart I, Personal Protective Equipment (general requirements)

[1]American Conference of Governmental Industrial Hygienists, Building D-5, 6500 Glenway Avenue, Cincinnati, Ohio 45211.
[2]American National Standards Institute, 11 West 42nd Street, New York, New York 10036.
[3]Compressed Gas Association, 1725 Jefferson Davis Highway, Arlington, Virginia 22202.
[4]Department of Transportation. The *Code of Federal Regulations* is available from the U.S. Government Printing Office, Washington, D.C. 20402.
[5]Environmental Protection Agency. The *Code of Federal Regulations* is available from the U.S. Government Printing Office, Washington, D.C. 20402.
[6]International Chamber of Shipping, International Association of Ports and Harbors, and the Oil Companies International Marine Forum, Portland House, Stag Place, London SW1E SBH, England.
[7]National Fire Protection Association, Batterymarch Park, Quincy, Massachusetts 02269.
[8]National Institute of Occupational Safety and Health, 125 Bakers Drive, Morgantown, West Virginia 26505.
[9]Occupational Safety and Health Administration, U.S. Department of Labor. The *Code of Federal Regulations* is available from the U.S. Government Printing Office, Washington, D.C. 20402.

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1910.134	Respiratory Protection
1910.146	Permit-required Confined Spaces (general industry)
1910.147	The Control of Hazardous Energy (lockout/tagout)
1910.251 et. seq.	Subpart Q, Welding, Cutting and Brazing (general requirements)
1910.1000 et. seq.	Subpart Z, Toxic and Hazardous Substances (PELs)
1910.1028	Benzene
1910.1025	Lead
1910.1200	Hazard Communication (includes labelling and MSDS provisions)
1926.62	Lead Exposure

SECTION 2—ADMINISTRATIVE CONTROLS

2.1 General

Each employer (owner, operator or contractor) shall develop and implement appropriate administrative controls and written plans for tank work from decommissioning through recommissioning that address the following:

a. Holding a pre-job discussion(s) with representatives of involved parties before their work commences.
b. Use of qualified person(s) at the job site to supervise and perform activities.
c. Training of personnel to use proper precautions and procedures.
d. Use of adequate safeguards to control hazards for protecting personnel and equipment.
e. Use of permits, such as cold work, entry, and hot work permits.
f. Use of appropriate ventilation procedures.
g. Testing by a qualified person of the tank atmosphere before and, as necessary, during entry to determine concentration of oxygen, flammable vapor, and toxic substances.
h. Inspection of the tank for physical and other hazards.
i. An emergency plan.
j. Compliance with applicable company policy, procedures, and programs and applicable government regulations.
k. Evaluation of contractors and/or subcontractors.

2.2 Qualified Person

The employer shall ensure that a qualified person or persons are designated to perform and/or supervise tasks or activities in accordance with the employer's administrative controls. For example, the following duties shall be performed by a qualified person: atmospheric testing, issuing permits (cold, hot, entry), and attendant duties.

2.3 Training

Employers shall ensure that employees understand the potential hazards of the job and the requirements, precautions, and procedures for controlling hazards. Specific training requirements are included in federal regulations, including OSHA, EPA and DOT.

2.4 Contractors

Contractors hired to work in tanks shall understand the hazards, be properly equipped, use trained personnel, and, as a minimum, observe the requirements, precautions, and procedures described in this standard. The owner and/or operator of the facility shall advise the contractor of the known potential hazards associated with the tank and the surrounding area. The owner and/or operator and the contractor shall agree on the requirements, precautions, and procedures to be followed and on the responsibility for meeting the requirements of 2.1.

For all work activities undertaken by the contractor, the owner and/or operator shall evaluate the contractor's experience, performance, work procedures, and practices. For additional guidance, see API Recommended Practice 2220.

SECTION 3—STORAGE TANK HAZARDS

3.1 General

All tanks that have contained petroleum or petroleum products may present one or more of the following hazards during some phase of preparing the tank for entry and cleaning:

a. Fires and explosions.
b. Oxygen deficiency or enrichment.
c. Toxic substances (liquids, vapors, fumes, and dust).
d. Physical and other hazards.

A more detailed description of potential storage tank hazards is provided in Appendix A.

3.2 Fire and Explosion Hazards

A fire occurs when the proper proportion of flammable vapor and air are exposed to an ignition source. The potential for a fire within or outside a tank can exist during all phases of tank work from decommissioning through recommissioning. Depending upon a number of factors, including

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tank design and integrity, air/fuel ratio, and the number of open vents, a fire inside a tank may cause an explosion due to expanding gases. During tank work, ignition sources shall be controlled to prevent possible ignition of flammable vapors. The atmosphere in the tank shall be tested (see Sections 5 and 6).

3.3 Oxygen Deficiency and Enrichment

An oxygen deficiency can exist in a tank for a number of reasons. The vapors of some hydrocarbons stored in atmospheric and low pressure (15 pounds per square inch gauge) tanks are heavier than air and will displace or dilute air (oxygen). If a tank has been closed, oxidation (rust) can deplete the tank's oxygen content. A tank that is maintained under an inert gas will be almost devoid of oxygen. Before workers enter a tank that has contained petroleum or petroleum products (or a closed and currently empty tank), the tank shall be tested for oxygen content and the atmospheric conditions evaluated (see Sections 5 and 6). While workers are in the tank, testing for oxygen shall be conducted, as necessary, to ensure that the oxygen content has not changed. If a tank has been maintained under an inert atmosphere, extreme caution must be exercised since the tank may be in an IDLH condition (see Section 6).

An oxygen enrichment hazard in which the oxygen content is over 23.5% should not normally occur during the cleaning of stationary storage tanks that have contained petroleum or petroleum products unless there are unusual or extraordinary circumstances. Any instance of oxygen enrichment within a tank above that of the ambient air external to the tank should be investigated. Occasionally, oxygen bottles or cylinders used during tank work may leak and provide a source of oxygen enrichment.

3.4 Toxic Substances

Toxic substances can cause irritation, injury, immediate or delayed illness, or death depending upon the characteristics of the substances, the concentration of the substances, and the duration of exposure. The possible routes of exposure are inhalation, skin or eye absorption, injection, or ingestion, depending upon the characteristics of the substance and the nature of the exposure.

As an example, hydrogen sulfide is an extremely toxic and flammable gas that occurs in the production, storage, and refining of sour crude oil and crude oil fractions. Tanks containing sour crude oil and crude oil fractions must be handled with extreme caution. Safeguards for prevention of exposure to hydrogen sulfide vapors are mandatory. Sweet crude oil and its components will normally not present a hydrogen sulfide exposure hazard unless there is or has been oil-reducing or sulfate-reducing bacteria in the tank. Precautions and procedures as noted in Appendix A should be observed. Also, leaded gasoline contains organic lead compounds, tetraethyl lead (TEL) and/or tetramethyl (TML) lead, which are highly toxic. Inorganic lead may also be present in paints. Precautions and procedures as noted in Appendixes A and D should be observed. Hydrocarbons may include components such as benzene that have been determined to have cancer-producing potential.

Information for the materials contained in the tank shall be obtained and reviewed by an appropriate qualified person before an entry permit is issued. Appropriate MSDSs shall be reviewed. Requirements under OSHA 29 CFR 1910.1200 should also be reviewed.

3.5 Physical and Other Hazards

Physical and other hazards may be present or created during tank work. Physical hazards may be a function of a tank's design or its physical condition, such as a corroded roof that is unsafe to walk on or work under. The internal structural components of a tank, such as heating coils, tank mixers, or sumps, may become obstacles for workers during tank work causing trips or falls. Workers inside the tank may encounter heat stress during warm or hot weather. Some crude oil may contain naturally occurring radioactive materials (NORM) (see API Bulletin E2). Potential physical and other hazards shall be evaluated.

SECTION 4—PREPARING THE TANK FOR ENTRY AND CLEANING

4.1 Initial Preparation

Hazards encountered while entering and cleaning tanks can be controlled by pre-planning the job in accordance with administrative controls (see Section 2). The essential elements for preparing a tank for safe entry and cleaning include the following:

a. A review of tank configuration, piping, and site layout. Review drawings and sketches, if available.

b. A review of information concerning the products contained in the tank to be entered and in surrounding tanks, including a review of applicable MSDSs or equivalent information.

c. Preparation of a safety and work plan for tank isolation, entry, and cleaning.

Additional guidance for specific types of tanks and specific products is provided in Appendix C.

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4.2 Emptying the Tank

The tank shall be emptied of all recoverable product through the tank discharge nozzle and line or other suitable connections. The unrecoverable product that remains below the tank discharge nozzle shall be removed using procedures established by the employer in the safety and work plan. Suggested methods for emptying the tank are included in Appendix E.

4.3 Isolating the Tank

The tank shall be isolated from all lines connected to the tank by using blanks and blinds or using double block and bleed valves, all of which shall be as close as possible to the tank. If the tank valve or line is to be removed and the tank has cathodic protection proceed as follows: (1) the cathodic protection should be turned off, (2) a bond wire should be installed from the tank to the lines, (3) the valve or line should be disconnected and removed, and (4) the bond wire should be removed. The lockout/tagout requirements of OSHA 29 CFR 1910.147 should be reviewed when planning tank isolation.

4.4 Control of Ignition Sources

Before any work is done that might involve the release of flammable vapors, ignition sources in the affected area shall be controlled.

4.4.1 MOTORIZED EQUIPMENT

Use of motorized equipment, such as automobiles, trucks, forklifts, welding machines, and compressors, shall be limited to safe areas by posting signs and/or barricading access, as necessary. See Appendix C for additional guidance.

4.4.2 ELECTRICAL TANK EQUIPMENT

Tank equipment such as in-tank mixers and instrumentation shall be disconnected and locked and/or tagged out.

4.4.3 ELECTRICAL MAINTENANCE EQUIPMENT

Equipment such as lights and motors used throughout the tank cleaning job shall meet the requirements of NFPA 70 for a Class I, Division 1, Group D (or greater) location unless other types of electrical equipment are specifically authorized under a job site procedure or work permit.

4.4.4 ELECTRICAL BONDING

The metallic parts of ventilating equipment, hoses used for washing the tank, and vacuum truck hoses shall be electrically bonded to the tank. Tank bond and ground cables and clamps shall be inspected for bonding and grounding integrity.

4.4.5 LIGHTNING AND THUNDER STORMS

When an electrical storm is in progress or threatening in the immediate area, all work inside and outside the tank shall cease and workers shall leave the tank and move to a designated safe place.

4.4.6 PYROPHORIC DEPOSITS

Precautions shall be taken to prevent pyrophoric deposits (often found in tanks containing sour petroleum or petroleum products) from becoming an ignition source. For guidance, see Appendix C.

4.4.7 IGNITION SOURCES FROM COMPRESSED AIR

If a compressor is used to pump air (non-breathing) into a tank to provide ventilation it shall be equipped with filters to remove moisture, scale, rust, and oil because moisture and particles in the air stream can generate electricity and become a source of static ignition.

4.5 Permit System

All work performed outside and inside the tank shall be done according to the work permit system established by the employer (see Section 2 and Section 8).

4.6 Vapor Freeing

The tank shall be freed of flammable and hazardous vapors by using appropriate methods that will safely ventilate the tank. Various methods for ventilating tanks are described in Appendix B. During vapor freeing, environmental concerns or regulations may also require additional steps to reduce atmospheric vapor emissions.

The employer shall determine the personal protective equipment and vapor concentrations at which workers may safely begin cleaning the tank from outside the manway without entering the tank. Because this process may require that workers' hands and arms break the plane of the manway, it may be necessary to issue an entry permit (see Section 6 for evaluating the hazards of tank entry).

4.7 Atmospheric Testing

Procedures for testing of the tank atmosphere shall be established prior to the initiation of tank cleaning. Atmospheric testing shall be conducted as specified in Section 5.

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SECTION 5—TESTING THE TANK ATMOSPHERE

5.1 Atmospheric Testing Equipment

For work outside and inside the tank, the tank atmosphere shall be tested or sampled with equipment that is intrinsically safe or approved for use in Class I, Division 1, Group D (or greater) locations. Manufacturers' recommendations, including calibration procedures, shall be followed. Documentation of instrument calibration shall be maintained for each instrument to ensure that a calibration program is in place. A record of the most recent calibration shall be maintained. Refer to appropriate MSDSs or equivalent information to determine substances contained in specific products for which the atmosphere shall be tested, using specialized toxic substance indicators (see 5.1.3). Additional guidance on the use of analyzers is provided in Appendix F.

5.1.1 OXYGEN ANALYZERS

The tank atmosphere shall be tested for oxygen content so that tank entry conditions can be evaluated as described in Section 6.

Oxygen analyzers are available as separate instruments or as part of a combination instrument.

5.1.2 FLAMMABLE VAPOR ANALYZERS

The tank atmosphere shall be tested for flammable vapor concentration so that tank entry conditions can be evaluated as described in Section 6.

Flammable vapor analyzers normally measure flammable vapor concentration as a percentage of the lower flammable limit, with 100 percent being the lower flammable limit. The analyzer shall be calibrated using a suitable gas of known concentration as specified or recommended by the instrument manufacturer.

See Appendix F for additional guidance concerning factors that may affect the accuracy of analyzers.

5.1.3 TOXIC SUBSTANCE INDICATORS

The tank atmosphere shall be tested for the presence of toxic substances as necessary so that tank entry conditions can be evaluated as described in Section 6.

A variety of direct-reading detector tubes and instruments are available for use in measuring low concentrations of hydrocarbon vapors, such as benzene and hydrogen sulfide. The accuracy and response times of these indicators vary greatly. The user shall understand these and other limitations of detector tubes and other instruments when used.

Lead-in-air analyzers or test kits[10], designed for measuring concentrations of tetraethyl and/or tetramethyl lead vapors, shall be used for determining organic lead-in-air concentrations in tanks that have contained leaded gasoline. Detector tubes or other instruments do not specifically detect organic lead vapors and shall not be used. Gas chromatographs may also be used. See Appendix D for guidance on performing lead-in-air tests.

5.2 Testing the Atmosphere

A qualified person shall determine the appropriate atmospheric tests that must be conducted. Such determination is based on a number of factors including the petroleum or petroleum product(s) that have been contained in the tank and the work to be performed.

A qualified person who is trained in the proper use and care of analyzers and other test equipment and in proper sampling procedures shall measure the atmospheric concentrations of oxygen, flammable vapors, and toxic materials outside and inside the tank.

The qualified person performing the test shall verify that the instrument(s) is in working condition and within calibration.

Most flammable vapor analyzers work on a design principle that depends on the presence of adequate oxygen levels to obtain accurate readings. For this reason, initial and subsequent tests inside the tank shall be conducted in the following order:

a. Oxygen content.
b. Flammable vapors.
c. Toxic vapors.

This order of testing is specifically mandated in OSHA 29 CFR 1910.146.

After initial tests indicate that workers may enter the tank, atmospheric testing shall be conducted, as required, to ensure the safety of workers inside the tank.

Tests for organic lead-in-air in storage tanks which have contained, or may have previously contained, leaded gasoline shall be performed after the tank has been thoroughly cleaned and dry. If tests are performed before or during the cleaning, the results can be erroneous and misleading. Appropriate personal protection equipment (see Section 7) is necessary until a tank is determined to be lead-hazard-free. There is no instrument that will detect or determine if the tank is or has been in leaded gasoline service prior to cleaning and drying. Therefore, unless the complete lifetime tank history is known, it should be assumed that the tank has been in leaded service unless tests prove otherwise. See D.4 for additional guidance.

[10]At the time of publication, there are only two known U.S. manufacturers of lead-in-air analyzers/test kits for measuring organic lead concentrations in the atmosphere of leaded-gasoline storage tanks: Core Laboratories, 19 Roszel Road, Princeton, New Jersey 08540; and Sensidyne, 16333 Bay Vista Drive, Clearwater, Florida 34620. API is concerned that the availability of these analyzers is diminishing rapidly. If other manufacturers, either domestic or international, provide lead-in-air analyzers, please contact API (Health and Environmental Affairs Department, Safe and Fire Protection unit) so that the information can be made available to potential users.

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5.2.1 TESTING THE ATMOSPHERE AROUND THE EXTERIOR OF THE TANK DURING VAPOR FREEING

The atmosphere around the tank shall be tested after the completion of preparatory steps and during the ventilation process to determine when and how much activity can take place around the tank. The ventilation equipment should be in place and operating. Tests shall be taken at a point where vapors are being exhausted from the tank and in the area around the tank to ensure that flammable and toxic vapors are not present above acceptable or permissible levels.

5.2.2 TESTING THE INTERNAL TANK ATMOSPHERE TO ESTABLISH ENTRY CONDITIONS

The internal tank atmosphere shall be tested to establish that permissible entry conditions exist throughout the tank (as specified in Section 6). Ventilating equipment should be shut-down for 15 minutes (based on industry experience) to allow the tank atmosphere to reach equilibrium conditions before testing the interior. The qualified person performing the test shall wear fresh air supplied or self-contained breathing apparatus (SCBA), positive-pressure full facepiece respiratory equipment, and appropriate protective clothing when entering the tank for testing purposes.

Care shall be taken by the qualified person who performs the tests to ensure that the results represent the true condition of the tank so that a decision can be made by the qualified person in charge of the work that personnel may safely enter the tank.

5.3 Monitoring Tank Atmosphere During Work

Atmospheric testing shall be conducted, as necessary, to protect workers while in the tank and to ensure that desired atmospheric conditions are maintained. Also see 9.1.

5.4 Documentation

The results of atmospheric testing, the time of the testing, and the signature of the person who performed the tests shall be recorded on or attached to the entry permit.

SECTION 6—HAZARD ASSESSMENT FOR ENTRY PERMITS

6.1 General

An entry permit shall be issued by a qualified person for *all* entries into a tank.

Conditions and precautionary measures for entry will be dependent upon the potential hazards associated with the entry, atmospheric test results, other physical hazards, and structural conditions and constraints. Equally important is the ability to maintain atmospheric conditions throughout the entry period.

The conditions within the tank will determine the level of precautions necessary for entry. The entry level and precautions for entry shall be determined based on atmospheric and physical conditions. The levels of precaution are:

Entry with Special Precautions
Entry with Restrictions
Entry without Restrictions

Judgement must be applied in selecting the appropriate level of protection for entry. The conditions indicated under each level are reference points and good engineering practice. The employer shall identify and establish any prohibitive entry conditions, as necessary. The conditions shall be carefully monitored throughout the entry, as necessary, to ensure conditions remain within permitted entry levels. If conditions change during the course of entry, all work shall stop and workers shall exit the tank until the conditions can be reassessed.

The intent of this tiered approach is to ensure precautions consistent with, but more protective than, OSHA requirements found in OSHA 29 CFR 1910.146, while at the same time providing for entry into confined spaces requiring special precautions, including entry in space maintained under an inert gas or other conditions immediately dangerous to life and health (IDLH). Note, that the level posing the least risk during entry is preferred.

6.2 Entry with Special Precautions

An entry permit shall be issued by a qualified person for all entries into a tank.

Entry with special precautions covers any entry into an atmosphere that may expose employees to the risk of death, incapacitation, impairment of ability to self-rescue (that is, escape unaided from the permit space), serious injury, or acute illness.

If any one of the following conditions exist, the entry shall be classified as a entry with special precautions:

a. Oxygen content is less than 19.5%.

Note: It is a practice in the petroleum industry to prohibit entry at levels above 23.5% oxygen unless there are extraordinary circumstances and appropriate precautions are taken.

b. Flammable vapors are above 10% of the lower flammable limit for the product in the tank.

Note: It is a long-standing practice in the petroleum industry to prohibit entry at levels above 20% LFL unless there are extraordinary circumstances and appropriate precautions are taken.

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c. Atmospheric concentration of any toxic substance at a level which will impair the ability to self-rescue (for example, hydrogen sulfide, certain substances used in cleaning and repair).
d. Any other atmospheric condition immediately dangerous to life or health (IDLH).
e. Any physical condition that may result in the engulfment of workers.

Due to the inherent danger of this category, extraordinary precautions and procedures shall be followed, and prohibited entry limits shall be established by the employer.

This type of entry is subject to the full requirements of OSHA 29 CFR 1910.146 concerning permit-required confined spaces, which must be reviewed for additional requirements before entry is allowed (use of an attendant, authorized entrant, entry supervisor, rescue plan, retrieval systems, etc.). See also the requirements under OSHA 29 CFR 1910.134 requiring the use of standby personnel.

Appropriate respiratory and protective equipment shall be selected to meet the existing and potential atmospheric conditions within the tank and those necessary to perform the specific tasks or activities. See the requirements of 29 CFR 1910.134 and substance specific OSHA regulations. Tanks that have contained leaded gasoline and have not been declared lead hazard free pose unique hazards (see D.4).

6.3 Entry with Restrictions

An entry permit shall be issued by a qualified person for all entries into a tank.

Employees will only be allowed to enter the tank wearing the appropriate respiratory and protective equipment as indicated on the entry permit for the atmospheric conditions. If the oxygen content is above 19.5% and below 23.5% and either or both of the following conditions exist, the entry shall be classified as an entry with restrictions:

a. Flammable vapors are greater than zero (0) but equal to or less than 10% of the lower flammable limit for the product that was contained in the tank.
b. Atmospheric concentrations of toxic substances are above their PEL/TLV and below the IDLH and do not expose employees to the risk of death, incapacity, impairment of the ability to self-rescue, serious injury, or acute illnesses due to health effects.

If the oxygen percentage in the tank differs from the ambient oxygen percent in the air external to the tank, additional evaluations shall be conducted since the differential could indicate the presence of hazardous contaminants or an oxygen source (see Appendix F).

Appropriate respiratory and protective equipment shall be selected, as necessary, to meet the existing and potential atmospheric conditions within the tank and to allow the performance of specific tasks or activities. See the requirements of 29 CFR 1910.134 or substance specific OSHA regulations.

Documentation shall be maintained that atmospheric and physical conditions have been evaluated.

6.4 Entry Without Restrictions

An entry permit shall be issued by a qualified person for all entries into a tank.

Employees may be allowed to enter without restrictions (in other words, no respiratory or protective equipment is necessary for the existing or potential atmospheric conditions) if the tank meets all of the following conditions:

a. Oxygen content is essentially equal to ambient level, normally 20.9%.
b. Flammable vapors are essentially 0% LFL for the product formerly contained in the tank, using typical field instruments to determine the level of flammable vapors.
c. Toxic substances are at or below published PEL/TLV limits as established by the employer and local, state, or federal governments.

Note: Exposure monitoring may be required to ensure that conditions are maintained at or below PEL/TLV limits.

d. The tank has not contained leaded gasoline or, if it did, it was previously cleaned and classified as lead-hazard free [in that lead-in-air test conducted after cleaning and drying indicated that the lead-in-air concentration is equal to or below 0.075 milligrams of organic lead per cubic meter (2 micrograms per cubic foot of air)] (see A.4.2 and D.4).

Documentation shall be maintained that atmospheric and physical conditions have been evaluated.

6.5 Initial Condition Assessment

All tanks shall be considered to require entry with special precautions (OHSA permit-required confined space) until the results of tank atmospheric tests demonstrate otherwise. If tank entry is required in order to verify atmospheric tests conducted from the exterior of the tank, then the appropriate level of respiratory and protective equipment shall be worn to protect the tester from any potential hazards within the tank. This would normally include wearing a positive-pressure self-contained breathing apparatus or a combination of a supplied-air respirator with a self-contained breathing apparatus with a full facepiece (see ANSI Z88.2).

Any tank that has been cleaned but closed and inactive shall be retested for entry as though it were the initial test.

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SECTION 7—PERSONAL PROTECTIVE EQUIPMENT

7.1 Protective Clothing

Employees who enter a tank shall wear the appropriate protective clothing for the atmospheric conditions and for the task or activity to be performed. Some items to consider in making the selection are the following:

a. The need for impervious clothing to protect the skin and the consequences of using impervious clothing related to heat stress.
b. The use of light colored clothing to enhance worker visibility and to assist in the identification of contamination on clothing.
c. The need for flame resistant coveralls.
d. The capability of certain clothing materials to produce a static discharge.
e. The need for hard hats, gloves, boots, goggles, etc.

If an employee and/or his or her clothing becomes contaminated with hazardous substances (from the tank or materials being used), the appropriate response actions shall be taken to decontaminate the person (including proper personal hygiene and showering, if necessary) and to decontaminate and/or dispose of the clothing.

See OSHA 29 CFR 1910.132 and specific health standards under OSHA 29 CFR 1000 et seq.

7.2 Respiratory Protection

To control occupational injuries, illnesses, and diseases caused by breathing air contaminated with harmful substances that cannot otherwise be controlled, appropriate respirators and respiratory practices shall be used by employees who are involved in entry and cleaning of storage tanks.

OSHA standards for respiratory protection [OSHA 29 CFR 1910.134 and the respiratory requirements for specific chemicals, for example, benzene (OSHA 29 CFR 1910.1028)] should be consulted. Appropriate respirators shall be used in accordance with 7.2.1 through 7.2.7.

7.2.1 RESPIRATORS SHALL BE PROVIDED BY THE EMPLOYER

Respirators shall be provided by the employer and shall be used by the employee when such equipment is necessary to protect the health of the employee. The employer shall provide the respirators which are applicable and suitable for the purpose intended. The employer shall establish and maintain a respiratory protection program which includes the requirements listed in 7.2.3.

7.2.2 TRAINING

The employee shall be trained in the use of respiratory protective equipment and shall use the respiratory protection provided in accordance with instructions and training received.

7.2.3 RESPIRATORY PROTECTION PROGRAM

The employer (owner, operator, or contractor) shall establish and maintain a respiratory protection program in accordance with OSHA 29 CFR 1910.134, including the following:

a. Written standard operating procedures governing the selection and use of respirators shall be established.
b. Respirators shall be selected on the basis of hazards to which the worker is exposed.
c. The user shall be instructed and trained in the proper use of respirators and their limitations.
d. Respirators shall be regularly cleaned and disinfected.
e. Respirators shall be stored in a convenient, clean, and sanitary location.
f. Respirators used routinely shall be inspected during cleaning, and respirators for emergency use shall be inspected at least once a month and after each use.
g. Appropriate surveillance of work area conditions shall be maintained, and the degree of employee exposure or stress shall be monitored.
h. There shall be regular inspections and evaluations to determine the continued effectiveness of the program.
i. Persons shall not be assigned to tasks requiring the use of respirators unless it has been determined that these persons are physically able to perform the work and use the equipment.
j. The respirator furnished shall provide adequate respiratory protection against the particular hazard for which it is designed in accordance with standards established by competent authorities. In the United States, competent authorities include the U.S. Department of Labor Mine Safety and Health Administration and the U.S. Department of Health and Human Services, National Institute for Occupational Safety and Health.

7.2.4 CHECKING AND FIT TESTING

Every respirator wearer shall receive fitting instructions, including demonstration and practice in the correct way to wear the respirator, proper adjustment, and ways to determine if it fits properly. Respirators shall not be worn when conditions prevent a good face seal. To assure proper protection, the facepiece fit shall be checked by the wearer each time he puts on the respirator. The fit check shall not be confused with either a qualitative or quantitative fit test which shall be performed periodically.

7.2.5 BREATHING AIR QUALITY

Breathing air shall meet the requirements of the specification for Grade D breathing air as described in CGA G-7.1-1966. Breathing air may be supplied to respirators from cylinders or air compressors. A breathing air compressor

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shall be used or an appropriate purification system provided or both to ensure the quality of the breathing air.

7.2.6 LOCATION OF BREATHING AIR SUPPLY

Breathing air cylinders shall be properly located to protect the air-supply standby person from potential exposure to flammable or toxic vapors that are both internal and external to the tank. The air supply shall be secured to protect it from physical damage and contamination.

7.2.7 AIR-LINE HOSE AND COUPLINGS

Air-supply systems shall be examined periodically to ensure the integrity and good condition of the air-supply lines and hoses, including air-line couplings. The breathing air-supply lines and hoses shall have no use other than for breathing air supply. Air-line couplings shall be incompatible with outlets for other gas systems to prevent inadvertent servicing of the air-line respirator with nonrespirable gases or oxygen.

SECTION 8—ENTERING AND WORKING INSIDE THE TANK

8.1 General

In addition to the requirements in Section 8 for entering and working inside of tanks, Appendix D provides precautions and guidance for cleaning leaded gasoline storage tanks and for some specific tank types.

8.2 Entry Permit

Before entering a tank, a written entry permit shall be issued by a qualified person and posted at the tank entry point. The written permit shall include, as a minimum, the following:

a. The permit space to be entered.
b. The purpose of the entry.
c. The date and the authorized duration of the entry permit.
d. The individual, by name, currently serving as entry supervisor, with a space for the signature or initials of the entry supervisor who originally authorized entry.
e. The hazards of the permit space to be entered.
f. The measures used to isolate the permit space and to eliminate or control permit space hazards before entry.
g. The acceptable entry conditions.
h. The results of initial and periodic tests performed, accompanied by the names or initials of the testers and by an indication of when the tests were performed.
i. The rescue and emergency services that can be summoned and the means (such as the equipment to use and the numbers to call) for summoning those services, if not provided elsewhere at the worksite.
j. Equipment, such as personal protective equipment and testing equipment required.
k. Any other information or permits whose inclusion is necessary, given the circumstances of the particular confined space, in order to ensure employee safety.

Title 29 CFR 1910.146 includes additional requirements if the tank meets the criteria for an OSHA permit-required confined space (entry with special precautions).

The qualified person(s) who issues the permit shall be familiar with the configuration of the tank to be entered and the reasons for tank entry. The qualified person(s) issuing the permit shall verify that conditions necessary for safe entry and work inside the tank have been met.

The entry permit shall be revalidated as stated on the permit. If conditions change during work to cause the tank to become more hazardous, the tank shall be evacuated and conditions reevaluated prior to reentry.

CAUTION: A clean tank that is closed and/or inactive shall be retested and permitted before workers may enter. The tank may be void of oxygen due to rusting or product may have seeped into the tank from a variety of sources.

8.3 Attendant

If a tank is entered while under Special Precautions (OSHA permit-required confined space) (see Section 6), at least one attendant shall be provided to meet the requirements of this standard and 29 CFR 1910.146. Also see the requirements under 29 CFR 1910.134 concerning standby personnel and chemical specific health standards. The employer shall ensure that each attendant:

a. Knows the hazards that may be faced during entry, including information on the mode, signs or symptoms, or consequences of exposure.
b. Is aware of possible behavioral effects of hazard exposure in authorized entrants.
c. Continuously maintains an accurate count of authorized entrants in the permit space and ensures that the means used to identify authorized entrants accurately identifies who is in the permit space.
d. Remains outside the permit space during entry operations until relieved by another attendant.
e. Communicates with authorized entrants as necessary to monitor entrant status and to alert entrants of the need to evacuate the space.
f. Monitors activities inside and outside the space to determine if it is safe for entrants to remain in the space and orders the authorized entrants to evacuate the permit space immediately under any of the following conditions:
   1. If the attendant detects a prohibited condition.
   2. If the attendant detects the behavioral effects of hazard exposure in an authorized entrant.

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3. If the attendant detects a situation outside the space that could endanger the authorized entrants.
4. If the attendant cannot effectively and safely perform all attendant duties.

g. Summon rescue and other emergency services as soon as the attendant determines that authorized entrants may need assistance to escape from permit space hazards.

h. Takes the following actions when unauthorized persons approach or enter a permit space while entry is underway:

1. Warn the unauthorized persons that they must stay away from the permit space.
2. Advise the unauthorized persons that they must exit immediately if they have entered the permit space.
3. Inform the authorized entrants and the entry supervisor if unauthorized persons have entered the permit space.

i. Performs non-entry rescues as specified by the employer's rescue procedure.

j. Performs no duties that might interfere with the attendant's primary duty to monitor and protect the authorized entrants.

The need for an attendant during entry into spaces that are maintained under conditions as defined in 6.3 and 6.4 shall be evaluated by a qualified person, and if an attendant is used, the attendant's duties shall be specifically identified.

8.4 Emergency Plan

An emergency plan shall exist for summoning rescue personnel and other emergency responders. A plan shall also exist for environmental contingencies.

8.5 Continuous Forced Ventilation

Flammable and toxic vapors may be present if product or sludge remains in the tank, if the tank has a leaking bottom, or if the product has collected in columns, roof supports, sumps, pontoons, floating swing lines or low spots. When these conditions exist, forced ventilation shall continue throughout the tank cleaning operation even if the test results are acceptable. See Appendix B for additional guidance on ventilation.

8.6 Retrieval Systems or Methods

Retrieval systems or methods meeting the requirements of OSHA 29 CFR 1910.146 shall be used for entry with special precautions (OSHA permit-required confined space). The use of retrieval systems or methods shall be evaluated by a qualified person for entry in accordance with the levels of precaution in 6.3 and 6.4.

8.7 Sludge Disposal

Disposal of sludge from tank cleaning operations must comply with applicable federal, state and local regulations (see Appendix D).

SECTION 9—HOT WORK AND TANK REPAIRS

9.1 General

Hot work is defined as any work that produces enough energy to be a potential ignition source. Welding, burning, grinding, drilling, abrasive blasting, and using of internal combustion engines are examples of hot work.

If tank repairs involve hot work, the surfaces to be heated, including roofs, structurals, scaffold decking, and pontoons, shall be free of flammable vapor and liquid, ignitable scale deposits, waxes, and other deposits that may evolve vapors. Such deposits are sometimes hidden on the upper surfaces of roof rafters. Flammable vapor analyzers may not indicate the presence of these materials. Wooden roof supports that emit vapors may be easily ignited. Flammable and toxic liquids and vapors may enter through leaks in the tank bottom or may evolve from pockets that exist in spaces such as pipe-column roof supports, chambers, swing lines, pontoons or sumps. The application of heat may produce flammable or toxic vapors. Frequent tests shall be made to verify that the atmosphere in the tank is not in excess of 10 percent of the lower flammable limit and that the PEL/TLV for toxic substances are not exceeded. Continued ventilation should minimize any accumulation of flammable vapors emanating from such sources. Additionally, local ventilation (a flexible hose with an air mover) may be used in some cases, such as when welding.

Adequate precautions shall be taken against potential exposure to toxic or other harmful atmospheres. Toxic atmospheres may originate from welding fumes or from the material being welded or worked on. The tank may have been coated with lead-based paint on the outside and/or coated on the inside with a special material to prevent corrosion (see A.4.6 for guidance on the potential for exposure to inorganic lead). Another potential hazard is the possibility of an oxygen-enriched atmosphere from leaking hoses, torches, or cylinder valves when using oxy/acetylene/ propane burning equipment. The composition and quantity of fumes shall be determined by investigating the materials being welded or worked on (for example, composition of welding rods and any coating or paint). Any requirement for local exhaust ventilation, type of respiratory protection and/or exposure monitoring shall be specified on the hot work permit. Oxygen and other gas bottles shall not be taken into confined spaces. Hoses and torches associated with this equipment shall be disconnected or removed during extended work break periods exceeding 1 hour unless the tank atmosphere has been retested before work resumes.

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Surfaces of a tank that have been in contact with leaded gasoline shall be scraped down to bare metal over any area of at least 12 inches (30 centimeters) on each side of the area that might be heated excessively by welding or by other hot work operations. In lieu of scraping, a wire brush may be used as long as appropriate protective clothing, respiratory equipment, and a face shield are worn. As an alternative to cleaning down to bare metal, welders and others in the area may use appropriate supplied-air respiratory equipment.

Because repairs to tank bottoms may create a fire, an explosion, or breathing (for example, inerting) hazards refer to API Publications 2207 and 2009 for additional guidance.

9.2 Hot Work Permit

Hot work may be performed inside the tank if all the conditions and requirements in 6.3 or 6.4 and 9.1 are met and maintained. A hot work permit shall be issued by a qualified person before hot work begins. Where hot work is to be performed, a fire extinguisher and/or pressured fire hose in working order shall be available, and personnel shall be instructed in its use.

9.3 Other Work

If the tank complies with the requirements of 6.4 at the outset, additional work, such as spraying, painting, or installing liners, may require a higher level of protection. The type of equipment or respiratory apparatus that would be appropriate depends on the substances, concentrations, and circumstances involved and should be evaluated in planning the operation.

SECTION 10—RETURNING THE TANK TO SERVICE

10.1 General

When recommissioning a tank, planning and procedures similar to those used to remove the tank from service shall be considered. Specific procedures for recommissioning the tank shall be developed.

If major repairs to the tank are performed, certain requirements prior to recommissioning must be met (see API Standard 653) to ensure its mechanical integrity (for example, hydrostatic testing).

10.2 Tank Closing

Prior to the installation of manway covers, the tank shall be thoroughly inspected to ensure that all workers have exited the tank and that maintenance equipment and/or debris which may hinder tank operations has been removed. Once this inspection has been performed, the tank manways may be installed. For additional guidance on recommissioning see Appendix E.

10.3 Recommissioning Equipment

All piping and equipment which was deactivated or isolated in order to remove a tank from service shall be reactivated in a manner which will not create any adverse conditions at the tank or place any part of the recommissioning process in a hazardous condition. For additional guidance, see Appendix E.

10.4 Filling Tanks

Visual examination or other equivalent methods shall be used to inspect the tank upon initiation of filling and then periodically until the tank has reached the operating capacity.

If the product to be pumped into the empty tank has the potential to emit flammable vapors, the flow rate (velocity) into the tank shall be reduced to a rate that will minimize the creation of static electricity while the outlet of the fill line is exposed (in other words, not covered with liquid). This rate, or velocity, varies depending on the product but is usually considered to be around 3 feet per second for steel pipe (see API Recommended Practice 2003).

When petroleum or petroleum product is pumped into an empty floating-roof tank, the vapors occupying the space between the liquid surface and the floating roof are usually expelled from under the floating roof to the atmosphere above the floating roof. Due to the potential for a flammable condition to exist, this activity shall be reviewed and proper precautionary measures taken.

The precautions concerning pumping rates and prevention of splashing to prevent static ignition and to minimize vapor generation should be observed with an external floating roof tank and an internal floating-roof tank until the roofs are floating. Sampling or hand gauging should not be done during this period (see API Recommended Practice 2003).

In the case of an internal floating-roof tank, vapor may be expelled from the eave vents. Pumping to refloat the roof shall not be done during a lightning storm. Other sources of ignition shall not be permitted near the tank during roof refloating.

CAUTION: Personnel should not go onto the top of a floating roof, internal or external, for a period of at least 18 hours after refloating the roof, or until the vapor concentration above the floating roof is 10 percent or less of the lower flammable limit (see API Publication 2026) and the potential for exposure to toxic materials has been evaluated.

CAUTION: Fill rates should be significantly reduced as product nears the pontoons and/or lower deck surface. If the liquid rise is too fast, damage to the roof might result (especially those made of aluminum).

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APPENDIX A—DESCRIPTION OF PETROLEUM STORAGE TANK HAZARDS

A.1 General

All tanks regardless of their size or type of hydrocarbon stored may present one or more of the following hazards during some phase of the tank entry and work:

a. Fires or explosions.
b. Oxygen deficiency or enrichment.
c. Toxic substances (liquids, vapors, fumes, and dusts).
d. Physical or other hazards.

The employer and the person directing the job at the tank site must be fully familiar with these hazards and with the methods of controlling them, using the precautions and procedures described in this standard.

A.2 Fires and Explosions

A fire, and possibly an explosion, may occur when the proper proportions of flammable vapors and air are exposed to an ignition source. The three elements necessary for a fire are flammable vapor, air (oxygen), and an ignition source as shown in the familiar "fire-triangle" (see Figure A-1).

An explosion results from the ignition of a flammable vapor-air mixture that fills a tank. The resultant fire inside the tank may produce a rapid pressure increase that exceeds the design strength of the tank and causes it to rupture. (See NFPA *Fire Protection Handbook*.)

A fire or explosion will not occur if any one of the three elements is missing. This fundamental principle is extremely important for the prevention of fires and explosions during work in and around tanks. Where flammable liquid fuels exist, flammable vapors will also exist above the liquid. To make tanks safe for work, both the hydrocarbon liquid and its vapor should be removed from the tank. After they have been removed, the atmosphere in the tank should be tested (see Section 5). The elimination of flammable liquids and vapors is the primary method for fire prevention during tank work.

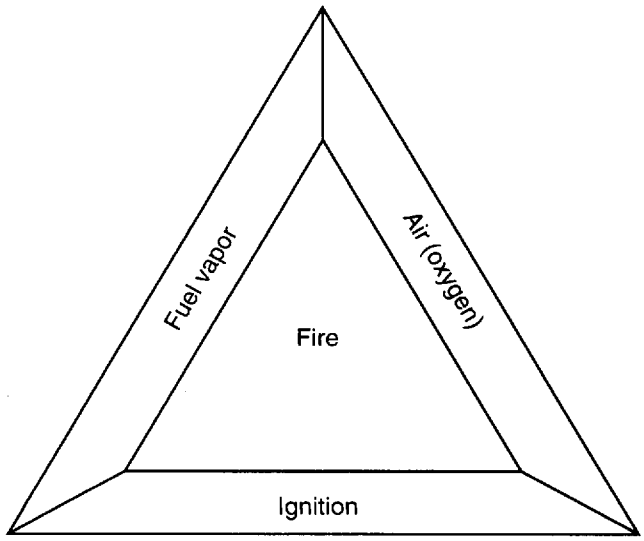


Figure A-1—The Fire Triangle

Eliminating ignition sources, especially during the vapor freeing process, is equally important. Elimination of ignition sources can be more difficult and less certain because ignition sources may go unrecognized or be remote. Unlike vapors and gases, ignition sources cannot be practically detected by an instrument. The vapors of some petroleum fuels, for example gasoline, are heavier than air and settle in low places. Vapors can spill from a tank and can flow like water. They can travel considerable distances and encounter a remote ignition source, for example, a car or truck engine, and be ignited. They can flash back to and into the tank where the vapors originated and cause a fire and explosion. For safe tank work, it is desirable to eliminate two of the three elements required for a fire, namely, flammable vapors and ignition sources.

The elimination of oxygen by inerting is another method of preventing fires but normally is not used for tank cleaning. The elimination of oxygen is difficult and costly and introduces another hazard, an oxygen-deficient atmosphere for workers (see A.3).

Mixtures of hydrocarbon vapor and air can be ignited only if the fuel-to-air ratio is within certain limits. For each hydrocarbon, there is a minimum concentration of vapor in air, below which a fire will not occur on contact with a source of ignition. There is also a maximum concentration of vapor in air, above which a fire will not occur. The boundary-line mixtures of vapor or gas in air that will, if ignited, propagate flame are known as the lower flammable limit (LFL) and upper flammable limit (UFL), respectively (see NFPA 325M). They are usually expressed as percents by volume of vapor or gas in air.

Different fuels have different flammable ranges. The flammable range of some common gases and vapors that may be present in and around tanks during entry, cleaning, and repair are shown in Table A-1.

Table A-1—Fuels and Their Flammable Ranges

Fuel	Flammable Range (volume percent in air)		
Hydrogen	4	to	75
Hydrogen Sulfide	4	to	46
Methane	5	to	15
Propane	2	to	9
Butane	2	to	8
Gasoline	1	to	7
Kerosene	1	to	5

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Information on the flammable range of materials can be found in NFPA 325M.

Figure A-2 shows the flammable range for gasoline vapor in air. Tank fires and explosions have occurred when workers were in a tank when the gasoline vapor/air mixture was thought to be in the rich range. It was mistakenly believed that a fire could not occur under such conditions. The extreme danger of this belief is that a rich atmosphere will ultimately dilute with air until the mixture is within the flammable range. Such dilution may occur at tank openings such as hatches, manways, or vents. Section 6 describes the conditions under which workers may enter a tank safely.

The importance of removing all recoverable hydrocarbon liquids and their vapors cannot be overemphasized. Even after a tank has been freed of vapor, flammable mixtures may be formed from any remaining residual liquid and sludge, or by the inadvertent entry of a liquid or vapor from an outside source.

Flammable vapor or liquid may enter a tank through unblinded lines or from leaks in the bottom of the tank. Flammable vapors may evolve within a supposedly empty and clean tank from sources that were overlooked. The most common sources are sludge, scale, hollow roof supports, unsealed sections of foam chambers, pontoons, heating coils, leaking bottoms, and internal wooden structures or other absorbent materials. Heat from the sun, from steam tracing, or from hot work may increase the tank vapor concentration. The tank atmosphere should be checked frequently for the presence of flammable vapors even though initial measurements indicated that the concentration was within acceptable limits (see Section 6).

Flammable vapor/air mixtures may be ignited by ignition sources, such as open flames, internal combustion engines, lightning, smoking, electrical shorts in worn or defective extension cords, and sparks. Spark sources include electric lamps, power tools, fixtures, switches, non-explosion-proof appliances, welding, and static electricity. Another source of ignition may be present in tanks that have been used for the storage of aromatic tars, asphalt, or sour stocks. The sulfur compounds in sour stock crude oil react with the steel of storage tank to form finely divided iron polysulfide deposits, which are pyrophoric on exposure to air. When these pyrophoric deposits come in contact with air, a heat-generating chemical reaction takes place. If the heat is not dissipated, spot temperatures high enough to ignite a flammable vapor can be reached (see B.5). Surfaces should be kept wetted during vapor freeing operations.

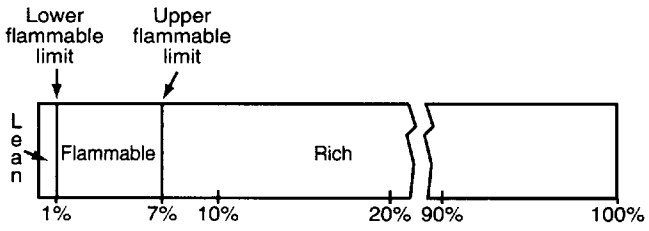


Figure A-2—Flammable Range for Gasoline Vapor in Air

Flammable deposits, such as condensed hydrocarbons or scale, may form on the underside of the tank roof or on rafters. The deposits can be ignited by cutting or welding operations on the roof. Such deposits should be wetted down and kept wet while hot work is being performed on the roof.

The cleaning of extremely large tanks may present special problems because of the tanks' very great volumes and large diameters. Special considerations should be given to ventilation and sludge removal. Special precautions and procedures, not included in this publication, should be established by the tank owner.

A.3 Oxygen Deficiency

Oxygen deficiency in a storage tank that has not been sufficiently ventilated can be a major threat to the workers inside the tank. Some flammable vapors are heavier than air. They may displace the air in the tank and make the atmosphere oxygen deficient. The atmosphere in the tank may become stratified, with the bottom portion of the tank having a lower oxygen concentration, the middle portion having some oxygen, and the upper portion being in the flammable range. A person entering an oxygen-deficient tank may be subjected to two potential hazards, fire and asphyxiation, and possibly to a third, toxic substances.

The oxygen content in a tank atmosphere must be determined with an oxygen analyzer before workers enter the tank. If the atmosphere contains less than 19.5 percent oxygen, it is oxygen-deficient, and appropriate respiratory equipment must be worn. It is always important to determine the reasons for an oxygen content that is not similar to normal/ambient air external to the tank (approximately 20.9%). See Appendix F for precautions concerning analyzer error when sampling oxygen deficient atmospheres.

Persons working in an oxygen-deficient atmosphere without appropriate respiratory protective equipment usually do not sense that the atmosphere is oxygen-deficient. There is little or no warning, and the effects are compounding, beginning with the loss of reasoning ability followed by unconsciousness. Death or brain damage can occur within minutes if the body is deprived of oxygen.

Air within a tank that has been closed and out of service may become oxygen-deficient as a result of the rusting of metal. Oxygen deficiency can also occur when inert gases, like nitrogen, are used or when inert welding gases are used.

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A.4 Toxic Substances

Information about health risks from particular substances should be obtained from the employer or the supplier of the substance. Material safety data sheets or equivalent information (available from the employer, manufacturer, or supplier) provides basic safety and health data for each hazardous substance produced or purchased by the employer. Government health, safety, and environmental agencies are additional sources of information.

Toxic substances can cause irritation, injury, acute or delayed illness, or death, depending on the characteristics of the substances, their concentration and nature and duration of exposure. Toxic substances can enter the body by inhalation, ingestion, skin and eye absorption, or injection. They can affect either the tissue at the point of contact or organs remote from the point of contact.

Irritants are substances that cause minor or transient but possibly painful injuries that heal without scars and produce no known aftereffects. Many hydrocarbons and polar solvents are irritants. Corrosives are substances that destroy tissue and leave permanent scars. Examples of corrosives are hydrofluoric acid, sulfuric acid, and caustics. Acutely toxic substances are those that by a single dose or short-term exposure may cause symptoms ranging from a simple headache or nausea to disablement or death. Hydrogen sulfide can be acutely toxic. Chronically toxic substances are those that may produce physiological impairment with long latency, such as cancer, or with gradual progression, such as pulmonary obstructive disease, or that may produce harmful reproductive effects.

Permissible exposure limits (PEL) as defined by OSHA, and *threshold limit values* (TLV) as defined by ACGIH (see section 1.4) are terms commonly used to describe the concentration in air of an airborne toxic substance below which, it is believed, nearly all workers may be repeatedly exposed without adverse affects. These terms are normally expressed in parts per million (ppm) per volume of air or in milligrams per cubic meter of air (mg/m3) in any 8-hour work shift of a 40-hour work week. MSDSs or equivalent information that provides toxicity information about exposure to specific substances may be obtained from the employer, manufacturer, or the supplier of materials.

Toxic substance hazards likely to be encountered in entry and cleaning storage tanks in the petroleum industry are listed below. The qualified person supervising the job should determine which toxic substance(s) are likely to be present in the tank to be entered.

a. Hydrogen sulfide.
b. Leaded gasoline.
c. Dusts.
d. Petroleum substances.
e. Welding fumes.
f. Lead-based paints.
g. Other chemical hazards.

A.4.1 HYDROGEN SULFIDE

Hydrogen sulfide (H_2S) is an extremely toxic flammable gas that is encountered in the production, storage, and processing of sour crude oil and sour crude oil fractions. It is colorless and in low concentrations has a foul, rotten-egg odor. Since hydrogen sulfide is heavier than air, it tends to collect in low places. The atmosphere in all tanks that have contained sour stocks should be considered toxic, and special precautions are necessary for safe tank entry. Hydrogen sulfide is typically eliminated by refining processes and is virtually absent from most finished products with the possible exception of some heavy fuel oils and some asphalts. In low concentration, it is detectable by its characteristic odor; however, the sense of smell cannot be relied on to warn of dangerous concentrations because instantaneous exposure to higher concentrations of the gas immediately paralyzes the sense of smell. Exposure to low concentrations for a longer period has a similar desensitizing effect. DO NOT DEPEND ON THE SENSE OF SMELL TO DETECT HYDROGEN SULFIDE.

In 1989, the Occupational Safety and Health Administration (OSHA) had prescribed updated exposure limits for several hundred chemicals, including new permissable exposure limits (PELs) for hydrogen sulfide of 10 parts per million time-weighted average (TWA) in an 8-hour work shift and 15 parts per million short term exposure limit (STEL) averaged over 15 minutes (54 Federal Register 2333; January 19, 1989). A federal court decision set aside the 1989 rule (58 Federal Register 35338; June 30, 1993). However, in the interest of safety and health and since some states may require them, the 1989 hydrogen sulfide limits should continue to be observed. Further, readers are urged to check the current editions of OSHA PELs and ACGIH TLVs, as well as current MSDS information.

The concentration of hydrogen sulfide in an atmosphere can be measured with various hydrogen sulfide indicators. The accuracy and response times of such indicators will vary. Users should understand these and other limitations of the instruments used.

Exposure to high enough concentrations of hydrogen sulfide causes death by paralysis of the respiratory system, and there is some indication that the presence of alcohol in the blood aggravates the effects of hydrogen sulfide in acute poisoning cases. In low doses, hydrogen sulfide is irritating to the eyes and respiratory tract. Repeated short exposures may lead to chronic irritation of the eyes, nose, and throat; however, the effects of such exposure are not cumulative, and the symptoms usually disappear soon after removal from the exposure. It should be understood that the sense of smell may be completely blocked by hydrogen sulfide. This impairment can result in a worker's failing to recognize the presence of dangerously high concentrations.

Hydrogen sulfide is a flammable gas that burns in air. Its flammable limits are 4 to 46 percent by volume in air, so pre-

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cautions against sources of ignition must be observed if it is present. A test for hydrogen sulfide should be made before any work is started on tanks that may have contained products containing hydrogen sulfide, such as sour stocks. The test should be repeated periodically.

A.4.2 LEADED GASOLINE

Leaded gasoline production and storage is very limited domestically; however, production is extensive internationally. Production of leaded gasoline is accomplished by adding an organic lead compound, a lead alkyl, to gasoline. The purpose of the lead alkyl is to raise the octane rating of gasoline. Lead alkyls, such as, tetraethyl lead (TEL) or tetramethyl lead (TML) or mixtures of the two, are toxic. When added to gasoline, the lead alkyl will create a potential organic lead hazard in tanks used to store leaded gasoline. Every storage tank that has stored leaded gasoline must be regarded as having an additional hazard, toxic organic lead. This hazard can exist in tanks that were formerly used to store leaded gasoline, but without cleaning were later switched to storage of a nonleaded product. The precautions and procedures in D.4 must be followed when entering leaded gasoline storage tanks or storage tanks that previously contained leaded gasoline and have not been determined to be lead-hazard-free.

Although the quantity of leaded gasoline being produced is diminishing in the United States, the vast number of tanks that present an organic lead hazard will probably continue to exist for some time, because in many cases, tanks were not cleaned and declared lead-hazard-free prior to the switch from the storage of leaded gasoline to the storage of nonleaded products. Persons responsible for tank entry should treat each gasoline storage tank as a leaded gasoline storage tank unless there is conclusive evidence that the tank was never used to store leaded gasoline or that it was cleaned and declared lead-hazard-free before the switch to storage of a nonleaded product. A lead-hazard-free tank is either a tank that was never used to store leaded gasoline or a leaded-gasoline storage tank that has been cleaned according to D.4, tested, and found to have an internal atmosphere below the current OSHA PEL of 0.075 milligrams of organic lead per cubic meter (2 micrograms per cubic foot). The operating record of a tank is the only reliable means of determining whether it was used for leaded gasoline. There is no analyzer or test method that can practically determine whether a tank has been used in the past to store leaded gasoline. Any attempt to test the tank before or during the cleaning process may result in unreliable information.

Lead alkyls are liquids that are totally miscible with gasoline. They are generally stable and will not settle out; there is essentially no decomposition of the lead alkyl in gasoline while in the tank. Lead alkyls have a vapor pressure lower than that of gasoline but are volatile enough to produce potentially lethal concentrations of organic lead vapors in the atmosphere of the tank. When cleaning a leaded-gasoline storage tank, the primary source of the organic lead hazard is in the sludge at the bottom of the tank. Sludge is a slurry of solids and liquids composed primarily of rust plus some water and entrained gasoline with organic lead. As the cleaning process progresses and the sludge is agitated, the more volatile gasoline components containing lead alkyls vaporize into the atmosphere.

Lead alkyls are toxic via all four routes of exposure: inhalation, skin and eye absorption, ingestion, and injection. Symptoms and effects vary depending on the degree of exposure and whether the toxic effect results from a single high-level exposure or from a relatively low-level exposure over a period of days. Exposures can be cumulative. Effects range from subtle to serious central nervous system disorders, psychological symptoms, and gastrointestinal dysfunction. Investigation shows that most organic lead poisonings during tank cleaning operations occurred because workers inside a leaded-gasoline storage tank did not wear air-supplied breathing equipment. Since the OSHA PEL for lead akyls also contains a skin designation, appropriate protective clothing and eye protection are necessary.

A.4.3 DUSTS

Dusts encountered in tank maintenance include those caused by the removal of rust, paint, and tank insulation. If dusts are present, appropriate respiratory protection and protective clothing should be worn. Examples of hazardous dusts are silica (from abrasive blasting), asbestos, lead, chromate, and zinc. See OSHA Title 29 CFR 1910.1000 and individual health standards (OSHA 29 CFR 1910.1000 and following) regarding mineral and nuisance dusts and the ACGIH booklet *Threshold Limit Values and Biological Exposure Indices* concerning various types of dusts.

A.4.4 PETROLEUM SUBSTANCES

Information about health risks from petroleum substances should be obtained from the employer or the supplier of the substance. MSDSs or equivalent information, available from the either the employer or the substance supplier, provides basic safety and health data for each hazardous substance produced or purchased by the employer. Government health, safety, and environmental agencies are additional sources of information.

The concentration of a hydrocarbon in most cases exceeds its permissible exposure limit or threshold limit value at levels well below 10% LEL (see OSHA Title 29 CFR 1910.1000 and individual health standards; also see Appendix F). For example, benzene is considered hazardous at very low concentrations in the range of 1-5 parts per million (ppm). Therefore, flammable vapor analyzers should not be used to measure airborne concentrations of toxic materials.

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Appropriate personal protective equipment should be worn according to recommendations in Sections 6 and 7 of this standard and applicable regulations.

The inhalation of high concentrations of hydrocarbon vapors can cause signs and symptoms of intoxication. These symptoms range from dizziness to excitement to unconsciousness; they are similar to those produced by alcohol or anesthetic gases. If such effects occur, the individual should be removed to fresh air. If the exposure is minor, breathing fresh air or inhaling pure oxygen may result in rapid recovery. If breathing has stopped, artificial respiration should be started promptly.

A.4.5 WELDING FUMES

The toxicity of welding fumes depends on their composition and concentration. The composition and quantity of fumes depend on the materials being welded, the compositions of the rods and of any coatings or paints affected, the process being used, and the circumstances of use. Toxic fumes are generated during welding on metals coated with or containing alloys of lead, zinc, cadmium, beryllium, and certain other metals. Some paints may produce toxic fumes when heated with a welding torch. The potential health effects of welding fumes vary in type and severity depending on these factors; some effects can be extremely serious. Detailed information on welding-fume toxicity and on appropriate protective measures, such as respirators, can be obtained from the employer and by consulting the ACGIH booklet, *Threshold Limit Values and Biological Exposure Indices,* as well as, OSHA Title 29 CFR 1910.251, Subpart Q, Welding, Cutting, and Brazing, and 1910.1000, Subpart Z, Toxic and Hazardous Substances.

A.4.6 LEAD-BASED PAINTS

Inorganic lead was once a common component of paints and coatings. Inorganic lead is a systemic, cumulative poison that can enter the body by inhalation of dust, fume, or vapor or by ingestion via eating, drinking, or smoking on the job or by eye contact. As the hazards of lead have become known, many of its uses have been discontinued. However, due to lead-based paint's resistance to rust and corrosion, it is still used for many industrial applications.

Significant lead exposure can occur during removal of paint from surfaces previously coated with lead-based paint, such as during the renovation or demolition of storage tanks. Operations that commonly generate lead dust and fumes include flame-torch cutting, welding or grinding of lead-based painted surfaces during dismantling and demolition work, abrasive blasting of lead-based painted structures, and using torches or heat guns, or sanding of lead-based painted surfaces during renovation work.

Cummulative exposure to lead may result in damage to the blood, nervous system, kidneys, bones, heart, and reproductive system. Lead may be stored in bone for many decades. Symptoms vary and include loss of appetite, constipation, and colicky abdominal pain. Nervous system complaints include headache and irritability.

Prior to commencing work (including demolition) on a tank, a written lead abatement program shall be developed for the safe removal and handling of lead-based paint. This program shall include procedures for the use of appropriate engineering controls, work practices, and personal protective equipment to keep exposures below permissible exposure limits. In addition, the program shall include procedures for training of workers, exposure assessment, and medical surveillance requirements. OSHA requirements regarding inorganic lead are found in OSHA 29 CFR 1910.1025 and 1926.62.

A.4.7 OTHER CHEMICAL HAZARDS

Additional information is required when tanks containing materials such as acids, caustics, high-boiling aromatics, and toxics, including naturally occurring radioactive materials, not specifically referred to in this publication are to be entered and cleaned. The MSDS or equivalent information for the material should be consulted for hazard identification and appropriate precautions.

A.5 Heat Stress

Workers engaged in work in the interior of a tank can encounter substantial heat stress when ambient temperatures are high and/or when workers are wearing protective clothing or equipment. If wearing semi-impermeable or impermeable clothing, workers can encounter heat stress working in temperatures as low as 70°F. A tank can absorb enough heat from direct sunlight to cause a large temperature increase inside the tank. The humidity in the tank is likely to be high, which further adds to the heat load. Workers should be monitored closely for signs of heat stress, and heat stress prevention techniques should be employed.

A.6 Physical and Other Hazards

In addition to injury or property damage that may result from fires, explosions, asphyxiation, or toxic substances, additional hazards may emerge from the following situations:

a. Inadequate training of workers or incompetent supervision.
b. Structural failure of the tank shell, roof, roof support members, swing line cables, or other tank members.
c. Falls through corroded roofs or from scaffolds, stairs, and ladders.
d. Entering and walking on an internal floating roof (see Appendix D).
e. Tools or other objects dropped from overhead.
f. Tripping over hoses, pipes, tools, or equipment.
g. Slipping on wet or oily surfaces or colliding with objects in inadequately lighted interiors.

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h. Accidental discharge of steam, high pressure air, water or oil, either into the tank or against personnel working outside, due to the omission of piping blinds.
i. Failure to wear appropriate respiratory protection.
j. Insufficient, faulty, or contaminated personal protective equipment or clothing.
k. Use of improper or poorly maintained tools, especially electrical tools or equipment.
l. Failure to disconnect or make electrical or mechanical equipment inoperative.
m. Noise above acceptable levels (for example, the use of power tools inside a confined space) (see OSHA 29 CFR 1910.95).
n. Use of lighting that fails to meet Class I, Division 1, Group D (or higher depending upon the type product in the tank) requirements. The use of ordinary household-type lights has been a frequent source of ignition in fires and explosions that have occurred during tank maintenance.
o. Inadequate working space.
p. Working in and around a tank during bad weather (for example, lightning or extreme temperature which can cause heat stress).
q. Poor illumination and shadows.
r. Crude oil containing naturally occurring radioactive materials.
s. Sharp angles, welding lugs, etc. within the tank that can snag clothing or skin in addition to tangling or cutting retrieval lines.

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APPENDIX B—VAPOR FREEING THE TANK

B.1 General

Note: Certain local conditions or regulations may not allow venting to atmosphere. In these instances, vapor recovery or treating or burning units may be required. These situations are beyond the scope of and are not covered by this standard or its appendixes. Check applicable laws and regulations regarding this issue.

A tank that has contained a low flashpoint product will contain high concentrations of flammable vapors and be above the upper flammable limit. During the vapor freeing process, the tank atmosphere will pass into the flammable range and ultimately to a range below the lower flammable limit. In the initial stage of vapor freeing, while the tank still contains high concentrations of flammable vapor, maintenance work in the immediate area surrounding the tank and on the tank roof should be restricted. An important consideration is to prevent the vapors that are being discharged from the tank from causing hazardous conditions outside the tank. Since some hydrocarbon vapors are heavier than air, discharging the vapors at the top of the tank or at a high elevation provides for quick dispersion. This precaution prevents flammable vapors from settling at ground level and flowing to an ignition source, thus being ignited and flashing back into the tank.

The selection of an effective vapor freeing method depends on many factors: for example, the type of product stored in the tank and the size and type of tank such as a cone roof tank, an external or internal floating roof tank, or a low pressure storage tank. There are several methods of vapor freeing a tank, including mechanical ventilation, steam ventilation, inert gas purging, and natural ventilation. If it is a small tank and its integrity permits, displacement of vapors by water flooding may be considered.

B.2 Mechanical Ventilation

Mechanical ventilation is normally the preferred method if the tank design, location, and the product stored in the tank permit this type of ventilation. An important feature of this method is that it can, depending upon where and how the air mover is installed, discharge vapors at a high elevation, for example, near the roof line. Mechanical ventilation equipment normally consists of eductors or blowers (air movers) that are driven by electricity, air, or steam (see Figure B-1). The equipment must be suitable for classified locations, and the air mover must be electrically bonded to the tank (see 4.4.3 and C.6). If the equipment is driven electrically, the requirements of 4.4.2 and C.5 should be met; if it is driven by steam, the steam exhaust must not enter the tank unless the precautions of B.3 are observed. Air movers which have moving parts or can otherwise produce a spark or ignition source should not be used in applications where the tank vapors may be in the flammable range and will pass through the air mover. Eductors with no moving parts should be used in such applications. See Figure B-2 for an air mover located on a shell manway of a tank.

B.2.1 MECHANICALLY VAPOR FREEING CONE-ROOF TANKS

There are several methods for mechanically vapor-freeing a cone roof tank such as the following:

a. Install an eductor (no moving parts) of the venturi type on a roof manway. Vapor is discharged upwards as fresh air enters the tank through bottom shell manways (see Figure B-3). With the bottom shell manways still closed, the air eductor should be started at a low delivery rate to create a slight negative pressure inside the tank before the bottom shell manway is opened. This procedure establishes a pressure differential so that there will be no release of vapor at ground level when the bottom shell manway cover is removed.

CAUTION: Before the bottom shell manway is opened, the negative pressure inside the tank should be kept to a minimum to prevent shell and roof plates from buckling and causing major tank damage.

After the shell manway cover has been removed, the air eductor may be operated at full capacity. Other bottom shell manways may now be removed if the eductor has sufficient capacity to cause fresh air to enter all of the open manways. Vapors should not be allowed to escape at bottom shell manways.

b. On one of the roof manways, install an eductor with a flexible duct which is attached to the suction side, and which extends into the tank and ends near the floor of the tank. Another roof manway is opened, but the shell manways are kept closed. The heavy vapor is drawn from near the bottom, flows up the tube through the air mover, and is discharged upward. If only one roof manway is available, it is possible to use a smaller eductor provided with an annularly-open support from the manway to allow fresh air entry.

c. Place an air mover (may be a blower fan type with moving parts) on the bottom shell manway and force air into the tank, allowing the vapor-air mixture to escape through the roof manway. The roof manway should be kept closed until after the blower is installed in the shell manway. This precaution minimizes the escape of vapor at the bottom shell manway. The roof manway cover is then removed and the blower started immediately.

B.2.2 MECHANICALLY VAPOR FREEING EXTERNAL FLOATING ROOF TANKS

The methods for mechanically vapor-freeing open-top external floating roof tanks are as follows:

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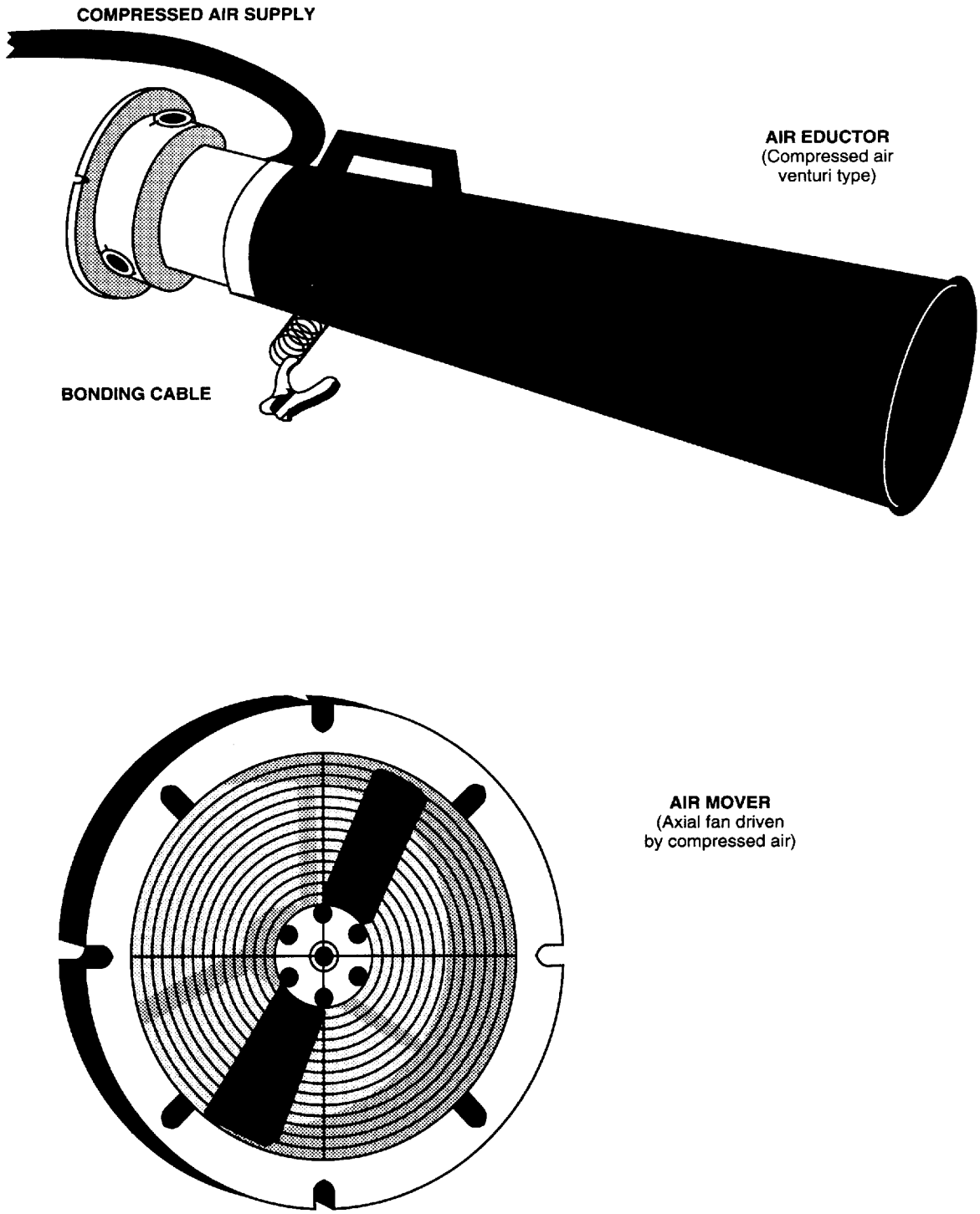
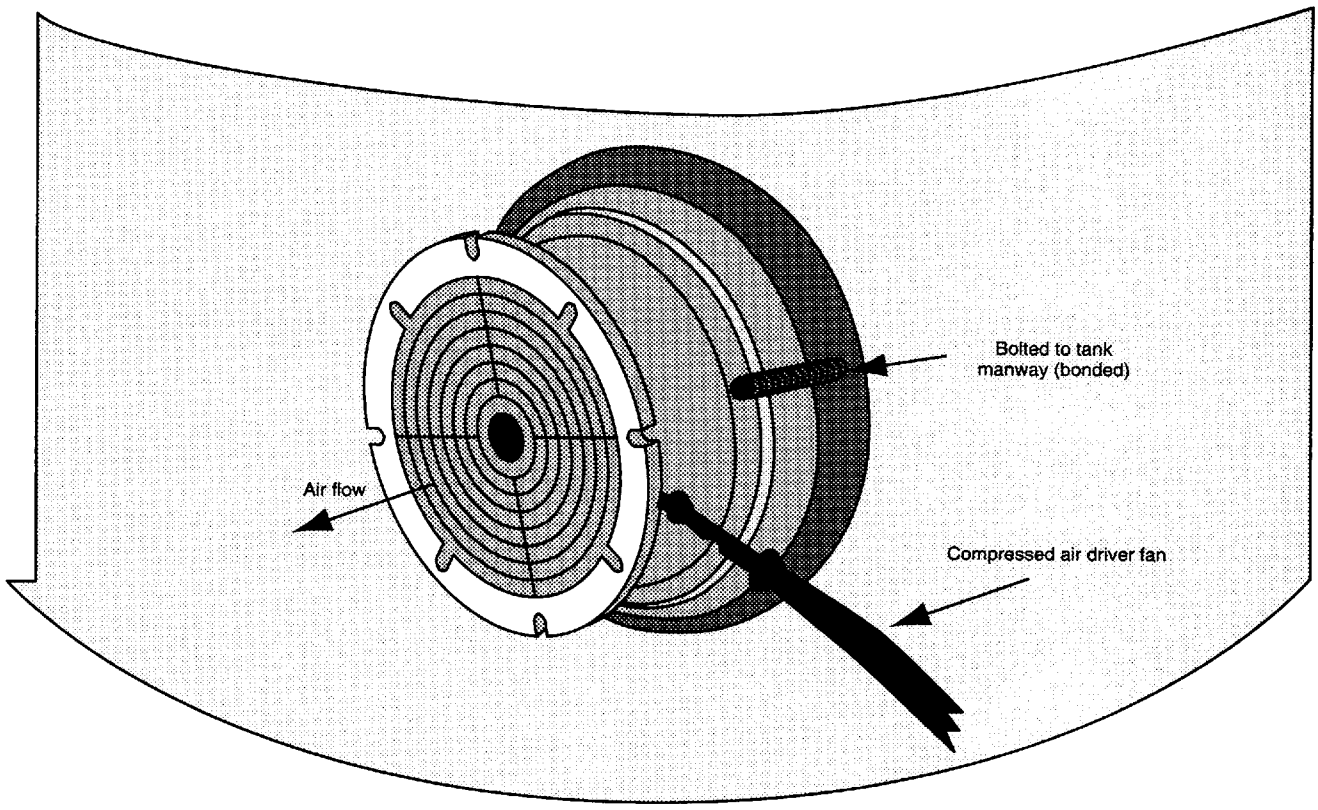


Figure B-1—Examples of Tank Mechanical Vapor-Freeing Equipment

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SAFETY PRECAUTION
Always be sure unit is connected to a suitable ground connection. Mounting on nonconductive adapters or free standing use requires a bonding cable.

Figure B-2—Air Mover Located on Shell Manway of Tank

a. The space above the floating roof may be mechanically ventilated by installing an air eductor at the top of the tank and connecting it to a duct that is lowered near the floating roof. This duct allows the vapor to be discharged at the top of the shell perimeter.

To vapor free the space below the floating roof, an air mover is installed on a bottom shell manway. The air mover is positioned on the manway so that it draws air through the tank. It is preferable to discharge into an elbow and a vertical duct to expel the vapor at as high a level as possible. After the air mover has been started at a low delivery rate, the remaining shell manways are opened, and the air mover is adjusted to full capacity.

b. The space below the floating roof may also be ventilated by installing an air eductor on the floating roof manway and connecting it to a vertical duct that discharges the vapor over the top of the tank shell. After the air eductor has been started, the shell manways are opened so that fresh air is drawn into the tank.

c. For small diameter tanks with only one shell manway, an air eductor can be installed in this manway. To ventilate under the roof, air is drawn in through the open piping attachment nozzles and through the floating roof's annular seal area. Floating roofs are usually equipped with a vent that opens when the roof is positioned on its legs. Air can be drawn in through this opening to provide additional circulation. The vapors should be discharged upwards through a duct. See Figure B-4.

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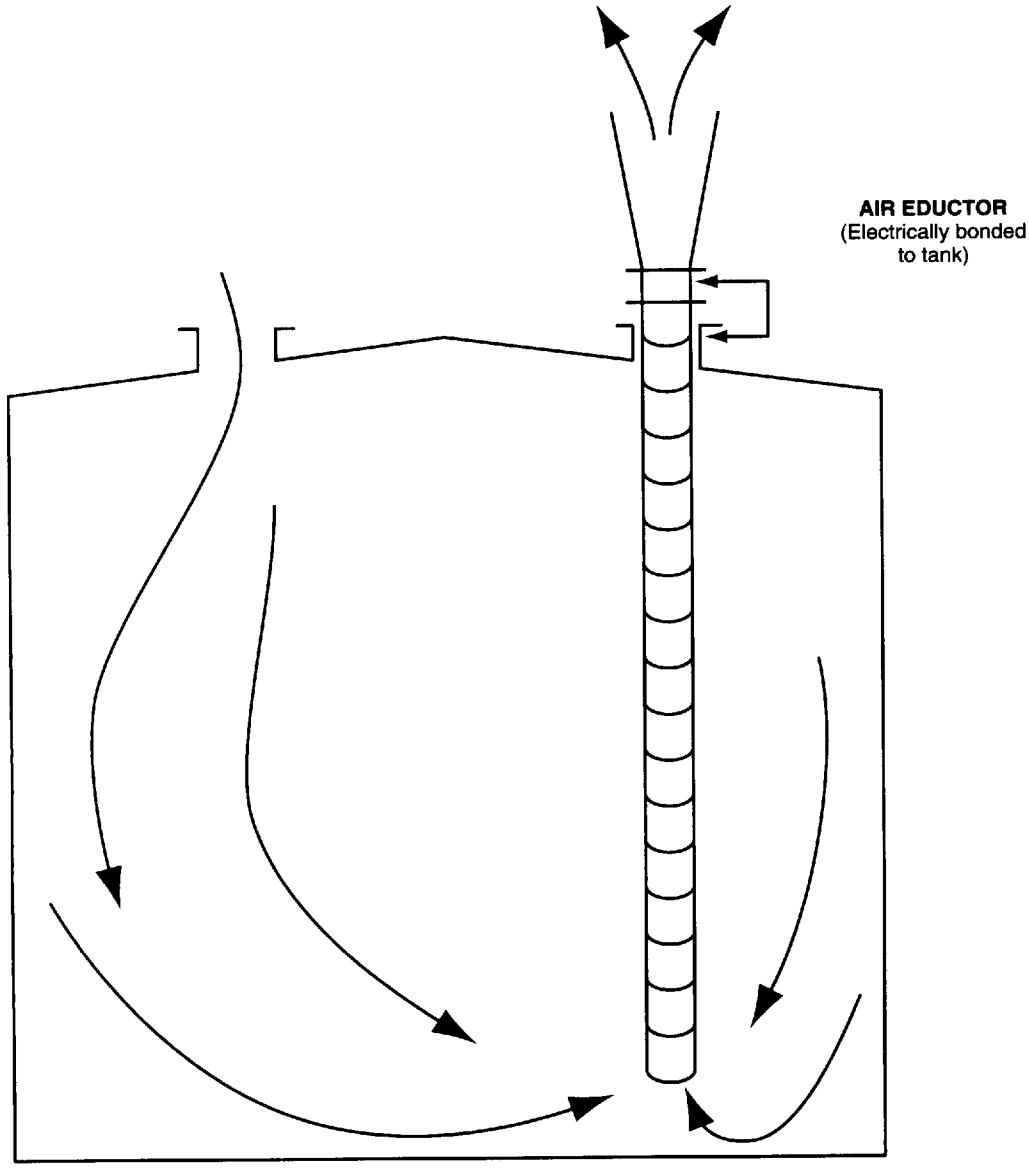


Figure B-3—Example of Vapor Freeing a Cone Roof Tank

B.2.3 MECHANICALLY VAPOR FREEING INTERNAL FLOATING ROOF TANKS

The methods for mechanically vapor-freeing areas above the internal floating roof are as follows:

a. Install on the fixed roof an air eductor that removes the vapor between the fixed roof and the floating roof. The vapor is discharged upward, and fresh air is drawn in through the shell eave vents. With the shell manways open, some vapor from beneath the floating roof will pass through the annular seal area and the column seals.

b. Install an air eductor on the fixed roof, and lower a duct to the floating roof. Vapor above the floating roof is exhausted as fresh air enters the shell eave vents.

Vapor freeing the space beneath the internal floating roof can be accomplished as described in B.2.2.

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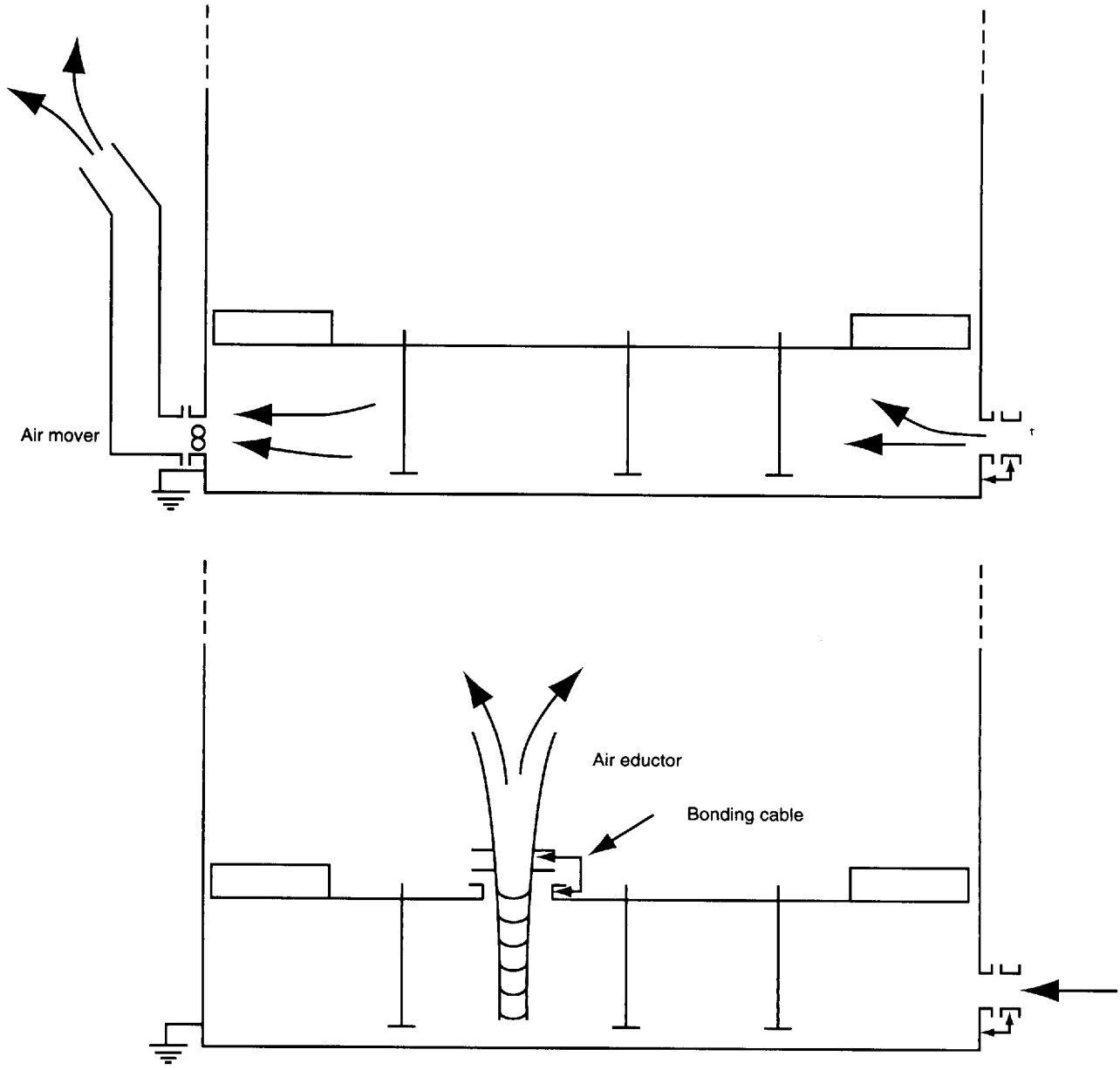


Figure B-4—Examples of Vapor Freeing an External Floating Roof Tank

B.3 Steam Ventilation

Steam ventilation may be advantageous in specific cases, but its use introduces some additional hazards. Steam ventilation should be avoided when suitable alternatives are available.

Steam jets are prolific generators of static charges on the nozzle and on objects that the steam impinges. These charges can cause sparks and may result in the ignition of flammable vapors. The pipe or nozzle through which the steam discharges must be electrically bonded to the tank; however, this bonding will not prevent a static charge from being generated after the steam leaves the nozzle. A charge may accumulate on an electrically insulated object inside the tank and result in sparking. Therefore, conductive insulated objects subject to steam impingement or condensation should also be electrically bonded to the tank shell (see API

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Publication 2003 and NFPA 77). The tank itself should also be grounded.

To be effective, steam must be introduced at a rate high enough to raise the temperature throughout the interior of the tank to at least 170°F (77°C). Often, the available steam may be insufficient to do this, particularly during cold weather or on large tanks. If the temperature in the tank reaches equilibrium below 170°F (77°C), the steam will condense as fast as it is introduced, and no more vapor will be expelled.

Steam should be introduced through a connection near the bottom of the tank, and either the roof manway or the gauge hatch should be left open during the entire operation. This precaution avoids both the building of excessive pressure while steaming and the creation of a vacuum while cooling. If steam vapor freeing is used, sufficient vacuum venting must be provided to prevent vacuum formation within the tank as a result of rapid steam condensation.

Steam vapor freeing may be appropriate for small tanks having an initial vapor/air atmosphere that is known to be vapor rich (above the upper flammable limit). Injecting purge steam through an enclosed piping connection ensures that the tank atmosphere is displaced by steam and never goes through the flammable range. For this method to be effective in removing all vapor, the steam rate must be high enough to produce a good steam plume from the roof manway which should ideally be located opposite the steam inlet to ensure a complete purge of the tank's interior.

B.4 Natural Ventilation

Natural ventilation should be avoided if other methods are available because vapor may drift to a source of ignition. Natural ventilation simply entails removing the roof and shell manway covers and allowing the tank to stand idle until natural ventilation makes the tank vapor-free. It is a slow process. At least at the start, the dense vapor will flow from the shell manway. As the ventilation progresses, the flow of vapor may reverse and flow from the roof manway because of changes in the density of the vapor, in wind velocity, or in atmospheric temperature. If this flow change occurs, it will most likely be at the end of the operation. Because drift and concentration of vapor are unpredictable, only workers wearing proper protective clothing and equipment, should be allowed in the area to make vapor tests.

B.5 Control of Pyrophoric Deposits in Asphalt Aromatic Tar and Sour Stock

If a tank has contained asphalt, aromatic tars, or sour stock, pyrophoric deposits may exist in the tank. If these deposits are allowed to dry out and react with oxygen in the air, they may spontaneously generate heat and cause ignition. This source of ignition can be controlled by dissipating the heat to prevent a temperature rise until the atmosphere inside the tank is below the flammable range. This hazard can be controlled by continuous wetting of the inside surfaces of the tank.

A suggested procedure for the removal of pyrophoric deposits from shell and roof surfaces involves the following steps:

a. Steam or water-fog spray the tank until the interior surfaces are wet with condensate. Fog nozzles must be electrically bonded to the tank.

b. Install a high-capacity air mover in the roof or shell manway and one or more large spray/fog nozzles. Electrically bond the air mover and fog nozzles to the tank.

CAUTION: The spraying of water into tanks can give rise to electrostatically charged mist. This has been the cause of a number of ship tank explosions. Refer to the ICS, IAPH, OCIMF *International Safety Guide for Oil Tankers and Terminals* for guidance.

CAUTION: See B.3 and B.5 for precautionary measures when using steam and Section 4 and C.6 for electrical bonding requirements for the air mover and for the water, fog, or steam injection nozzles.

c. Without delay, turn on the fog nozzles and immediately thereafter, turn on the air mover. (The interior surfaces of the tank will be kept wet and the ingoing air will be thoroughly wetted by the moisture from the fog stream.) The air mover and fog nozzles must be operated without interruption until the tank is vapor-free.

d. With the air mover still in operation after the fog nozzle has been removed, knock down all loose scale with a high-pressure water stream.

CAUTION: The requirements of Sections 6 and 7 must be met before workers may enter the tank.

e. Proceed with the tank cleaning operation.

B.6 Inert Gas Purging

One means of vapor freeing a tank is to use an inert gas purge. If this method is used, additional precautions are necessary to ensure appropriate ventilation to remove the inert gas before entry and cleaning. Alternatively, work may be conducted after taking very special precautions appropriate for entry into a confined space under conditions immediately dangerous to life and health (IDLH) as outlined in 6.2. For example, open manways during inert gas purging could be fitted with a warning and/or restriction devices to prevent entry.

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APPENDIX C—CONTROL OF IGNITION SOURCES

C.1 General

Before any work is done that might involve the release of flammable vapors, sources of ignition such as smoking, welding, or grinding should be prohibited where flammable vapors may be present or may travel.

Unexpected sources of ignition occur. It is not sufficient just to eliminate conditions known to be a possible source of ignition at the tank site. Every effort must be made to avoid the release of vapors near ground level during ventilation and cleaning operations. Many hydrocarbon vapors are heavier than air and can travel along the ground for considerable distances and possibly encounter an ignition source.

The work area should be kept free of internal combustion engines and electrical equipment that is not properly classified until the tank is vapor free and the sludge has been removed. Electrical bonding of hose nozzles and ventilating equipment is most important. No work should be done if the direction of the wind might carry vapors into areas where they might produce a hazardous condition.

C.2 Vehicular Traffic and Vacuum Trucks

Internal combustion engines are ignition sources which must be controlled in the tank area. Roads should be barricaded or posted or both as necessary to control vehicle traffic into the diked area.

If a vacuum truck is used to remove sludge and waste from the tank, careful consideration must be given to its placement and to the truck's distance from the tank to ensure that flammable vapors will not reach the internal-combustion engine. The vacuum truck tank exhaust vapors should be discharged through a hose of adequate size and length downwind of the truck and away from the storage tank. The vacuum truck suction and discharge hoses must be electrically bonded and grounded (see API Publication 2219).

C.3 Lightning Storm

While a tank is open and being ventilated and/or being cleaned, all personnel should leave the tank cleaning site if an electrical storm is in progress or is threatening, because lightning is a potential ignition source (see 4.4.4).

C.4 Artificial Lights

The use of artificial lights in or around tanks [except dry-cell flashlights approved for Class I, Division 1, Group D (or higher) locations] should be avoided, if possible, when there is a potential of flammable vapors exceeding 10 percent of the lower flammable limit. When other than approved dry-cell lights must be used, the lights, extension cords, connectors, switches, and accessories must provide protection from electrical shock and must be provided with ground fault protection and must meet the requirements of NFPA 70 for a Class I, Division 1, Group D (or greater) location. Before use, such equipment should be thoroughly inspected to ensure that it will not be an ignition source or a shock hazard (see 4.4).

C.5 Motors

All motors and extension cords, connectors, switches, and accessories, such as those used to operate ventilating equipment, compressors, and pumps for removal of product and sludge should meet the requirements of NFPA 70 for a Class I, Division 1, Group D (or greater) location and afford the level of protection specified in 4.4.

C.6 Electrical Bonding

The metallic parts of ventilating equipment and duct work (see Appendix B) must be electrically bonded to the tank to avoid sparks. Vacuum truck hoses and hose nozzles used to dislodge and flush sludge to a water draw or pumpout connection, or hose nozzles used to inject product or water into the tank must be electrically bonded to the tank. If abrasive blasting is done, hose nozzles must be bonded to the tank (see API Publication 2027, Recommended Practice 2003 and 4.4). If ventilation is done by injecting steam, see bonding requirements in B.3.

C.7 Control of Pyrophoric Deposits

If a tank has contained asphalt, sour stocks or aromatic tars, pyrophoric deposits (iron sulfide) may exist in the tank. If these deposits are allowed to dry out and react with the oxygen in the air, they may spontaneously generate heat and cause ignition (see Appendix A). This dissipation source of ignition can be controlled by dissipating the heat to prevent a temperature rise until the vapor inside the tank is below the flammable range. This process may be accomplished by wetting all interior surfaces of the tank with water. The wetting dissipates the heat of reaction and also tends to isolate the pyrophoric deposits from oxygen present during ventilation. Continuous wetting of the inside surfaces of a tank can be accomplished by positioning hoses with fog nozzles at open manways. The fog nozzles must be electrically bonded to the tank shell (see 4.4.3, 4.4.4 and B.5).

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APPENDIX D—TANK CLEANING PRECAUTIONS AND PROCEDURES FOR SPECIFIC TANK DESIGNS AND PRODUCTS

D.1 General

Tank entry and cleaning precautions and procedures described in this standard and appendixes apply to all petroleum and petroleum product tanks that are within its scope. This appendix provides additional guidance for tank entry and cleaning requirements for the following types of tanks:

a. Cone roof tanks.
b. External and internal floating roof tanks
c. Leaded gasoline tanks

D.2 Cone Roof Tanks

The precautions and procedures as described in this standard and in Appendixes A, B, C, and applicable sections of D also apply to cone roof tanks. However, multi-million barrel cone roof tanks may present special problems due to their very great volumes and large diameters. Special considerations may have to be given to ventilation and sludge removal. Special precautions and procedures, not included in this publication, should be established by the tank owner/operator.

D.3 Cleaning External and Internal Floating Roof Tanks

The precautions and procedures described in this standard and appendixes apply to floating roof tanks.

D.3.1 FLOATING ROOF DESIGN

Personnel responsible for entry and cleaning of external and internal floating roof tanks should be familiar with those aspects of floating roof tanks and roof design that affect entry and cleaning activities. External floating roofs have steel-enclosed compartments for flotation; these compartments may be peripheral pontoons or a complete double deck. External steel pans have also been used. External tanks have no fixed roof over the floating roof.

Internal floating roofs employ several methods of providing buoyancy. Some manufacturers use aluminum tube pontoons. Others employ an inherently buoyant foam material enclosed in metal or plastic sheeting to form the deck. Some internal roofs are simple steel pans or pontoon types. Internal floating roofs have a fixed roof covering the tank that forms a vapor space between the floating and fixed roofs.

The seals that close the annular space between the floating roof and the shell may be flexible foam-filled bags, flexible wipers, or metallic shoes. Primary and secondary seals with an intermediate vapor space are often used on external floating roof tanks.

D.3.2 HYDROCARBON ENTRAPMENT

Any of these buoyant and sealing devices may retain hydrocarbon and leak after the tank appears to be vapor free and after satisfactory vapor indicator tests have been obtained. Hydrocarbons may also be trapped in the roof support legs.

Before cleaning begins on a floating-roof tank, knowledgeable personnel should review and be familiar with the roof design. In particular, the type of flotation and sealing devices used should be studied. Pontoons should be inspected. If they contain hydrocarbons, the hydrocarbons should be removed to minimize the potential of fire, oxygen, and toxicity hazards.

CAUTION: Liquids in pontoons may also create a physical hazard because the roof and roof legs are not designed to carry additional weight.

D.3.3 INFORMATION ON CONDITION OF FLOATING ROOFS

Available information on experience with leakage and retention of hydrocarbons in roofs of the type to be worked on should be reviewed. In addition, particularly in older tanks, information on the various services in which the tank has been used should be evaluated.

D.3.4 EFFECTS OF FLOATING ROOF DESIGN ON CLEANING ACTIVITIES

The restricted space under floating roofs resting on legs on the bottom of the tank should be taken into account in planning work and atmospheric testing. The height of the roof above the tank bottom may not permit personnel to walk erect.

Descent onto internal floating roofs and small external floating roofs is usually made by means of a vertical ladder. Caution must be exercised in using the ladder during gas testing and cleaning activities; a rapid exit should be carefully planned. The floating roofs of external floating roof tanks are usually well constructed and unless corroded or otherwise defective will support the weight of personnel. Many internal floating roofs are of light construction, and walking on the roof surface can be hazardous unless limited to areas designated for that purpose. A portable walkway is illustrated in API Publication 2026. Defective roof areas or panels which were not designed to support the weight of personnel may be difficult to detect because of the restricted visibility above the roof.

Plastic used in internal floating roofs may be combustible, and this characteristic should be considered, particularly if activities in the tank will involve hot work or other sources of ignition. Seals may also be made of combustible material or may be oil soaked.

D.3.5 DETERMINATION OF FLOATING ROOF CONDITION

Visual examination through available openings in the fixed roof should be made to determine, as well as possible,

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the condition of the floating roof. In the case of external floating roof tanks, the roof condition should be reasonably apparent from the top of the stairway.

D.3.6 DESCENT ONTO FLOATING ROOF

API Publication 2026 should be carefully reviewed before entry onto the roof.

D.3.7 POSITIONING ROOF SUPPORT LEGS

It is generally necessary to descend onto the floating roof while the tank is still in service in order to reposition the roof support legs so that the roof will land in the high roof position. All the precautions and procedures described in API Publication 2026 should be followed. When preplanning allows it, descent onto the floating roof should be made after the roof has been raised to the highest level which permits safe entry space. This timing of descent maximizes natural ventilation and minimizes climbing. The roof support legs and sleeves through the roof should be examined for corrosion or other defects to determine their ability to support the roof's weight. Leg positioning pins should be located in the high roof position with retaining cotter pins. This precaution will facilitate cleaning operations by supporting the roof at as high a position as possible above the tank bottom.

If the support legs are designed for only one support level or are already in the high roof position, entry for leg positioning will not be necessary.

D.3.8 VAPOR FREEING THE TANK

The precautions and procedures described in Section 4 and Appendixes B and C must be followed when pumping out and emptying the tank and when vapor freeing the tank.

CAUTION: Certain areas may release hydrocarbon liquid or vapor after initial vapor freeing.

Even though a tank has been tested and found safe for entry it will not necessarily remain in this condition, particularly in the case of floating roof tanks. With the passage of time, leakage of liquids or vapor from pontoons, buoyant foam panels, other flotation devices, linings, and seals may result in toxic or flammable vapors being present in the tank vapor space. A build-up of vapors may be more likely in internal floating roof tanks because of the possibility of liquid being retained in the envelope construction of flotation and sealing components. These tanks should be inspected for presence of liquid. If liquid is present, it should be removed before work is begun. Atmospheric monitoring is necessary prior to the start of each work period and at pre-determined intervals during work periods. Frequent vapor monitoring is recommended.

D.3.9 TESTING THE ATMOSPHERE ABOVE AND BELOW THE FLOATING ROOF

Testing requirements described in Section 5 should be followed. In the case of floating roof tanks, particularly the internal type, initial oxygen, flammable vapor, and toxic substances tests may be made with an instrument equipped with an extension device. This testing can be done from the top of the tank so that preliminary readings can be obtained without descending onto the internal roof. When obtaining the sample through an extension device, the instructions of the test instrument manufacturer should be carefully followed to ensure that accurate readings are obtained. For descent onto the tank roof for further testing, the tester should have appropriate respiratory protection and protective clothing according to the entry categories described in Section 6.

D.3.10 SAFETY AND RESCUE EQUIPMENT

The safety plan for tank work should include a means for timely rescue of workers from the tank.

Self-contained breathing apparatus should be immediately available at the tank for both internal and external floating roof tanks for rescue purposes by trained and qualified personnel. Consideration should be given to the need for the following equipment where descent is to be made onto internal floating roofs in service:

a. A lightweight material to spread the weight of personnel over a greater area of the floating roof.
b. A rescue toboggan, which will fit through the roof manway for lifting an incapacitated person out of the tank.
c. Approved harness safety belts and lifelines.
d. A davit which can be attached to the fixed roof manway of an internal floating roof tank for lifting the toboggan with an incapacitated person out of the tank. Drawings of a lightweight walking surface, rescue toboggan and davit appear in API Publication 2026.

D.3.11 CONDITION OF ROOF SUPPORT LEGS

Observation should be made to determine the condition of the legs supporting the floating roof. If the legs are corroded or otherwise defective, temporary roof support may be necessary while legs are being repaired and/or replaced.

D.3.12 RETURNING THE TANK TO SERVICE

When a flammable product is pumped into an empty floating roof tank, some of the vapors above the liquid will remain below the floating roof and some may be expelled to the space above the floating roof until the roof floats (see Section 10).

CAUTION: Vapor above or below the roof may be in the flammable range.

In the case of internal floating roof tanks, vapor may be expelled from the eave vents. Pumping to refloat the roof

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should not be done during a lightning storm or with electrical or mechanical ventilating equipment in service. Other sources of ignition should not be permitted in the vicinity of the tank during roof refloating.

The precautions concerning pumping rates and prevention of splashing to prevent static ignition and to minimize vapor generation should be observed with external and internal floating roof tanks until the roof is floating. Sampling or hand gaging should not be done during this period (see API Publication 2003).

Personnel should not go onto the top of internal floating roof tanks or on the floating roof of external floating roof tanks for a period of approximately 18 hours after refloating or until the vapor concentration above the roof is below 10 percent of the lower flammable limit and the atmosphere is safe to breathe.

D.4 Cleaning Leaded Gasoline Tanks

D.4.1 GENERAL

All tanks that are used to store leaded gasoline present a potential organic lead hazard to workers engaged in cleaning them. See Section 6 and Appendix A.4.2. Tanks that have at one time stored leaded gasoline but have been switched to unleaded gasoline storage without being cleaned present a potential organic lead hazard. Even if such a tank has been in unleaded product service for many years, the organic lead hazard can remain indefinitely, primarily in the sludge on the tank bottom.

After a leaded gasoline storage tank has been thoroughly cleaned and tested with a lead-in-air analyzer or other suitable means and determined to be lead-hazard free, the tank may then be treated as a nonleaded product storage tank except that such a tank should not be used for the storage of edibles or potables for animal or human consumption. If the tank is dismantled, the steel must not be used for any purpose that involves contact with edibles and/or potables (see API Publication 2202).

The safety precautions and procedures for cleaning leaded gasoline storage tanks include all the procedures for cleaning a nonleaded product tank plus the procedures in this appendix. The following is emphasized:

a. Workers inside the tank must wear positive pressure full facepiece, (supplied air or SCBA) respiratory equipment, and appropriate eye and skin protection throughout the tank cleaning process regardless of the flammable vapor and oxygen concentrations.
b. In addition to the personal protective clothing requirements described in Section 7 of this standard, workers should wear clean clothing from the skin out each day along with impervious boots and gloves. If clothing becomes contaminated, for example, with sludge, during work procedures, the worker should leave the tank, remove the clothing, shower, and put on clean clothing before resuming work.
c. After the tank has been cleaned and dried, the tank atmosphere should be tested with a lead-in-air analyzer or other suitable means (see D.4.2, D.4.3, and D.4.4).
d. Sludge from the tank must be disposed of according to applicable regulations (see D.4.4).
e. If maintenance work, including hot work, is to be done after the tank has been cleaned, API Publications 2202 and 2207 should be consulted.

D.4.2 LEAD-HAZARD-FREE TANK

A tank is described as lead-hazard-free when the following conditions are satisfied:

a. The tank has been cleaned and all sludge has been removed.
b. Loosely adherent materials have been removed from the part of the tank that has been in direct contact with sludge, normally the tank bottom and perhaps the side walls of the first shell ring.
c. The tank is essentially dry and free from liquid or puddles.
d. A lead-in-air test which must be performed *after* the tank has been cleaned and dried indicates the organic lead-in-air concentration is equal to or less than the permissible exposure limit.

The tank is *not* lead-hazard-free if, after cleaning, any of the following conditions prevail:

a. A lead-in-air test has not been made, or the test results exceed the permissible exposure limit.
b. Surfaces of the tank that were in contact with lead material are heated, for example, by cutting, burning, or welding, unless all rust, scale, and other foreign material have been removed to expose bare metal at least 12 inches (30 centimeters) on each side of the area that will be heated.
c. The tank contains absorbent materials that may release organic lead.
d. The tank has a leak source.

Frequent lead-in-air tests should be made to ensure safe conditions when repairs are made.

D.4.3 LEAD-IN-AIR ANALYSIS

The analysis used to measure the concentration of lead-in-air must measure organic lead, not inorganic lead (see 5.1.3). Manufacturer's operating instructions should be followed for instrumental analyzers or analytical methods consistent with OSHA/NIOSH protocol for measuring lead-in-air concentrations (see NIOSH *Manual of Analytical Methods*, Method 2533).

The lead-in-air analyzer should *not* be used before or during tank cleaning; the results may not reflect the true condition of the tank. The results of testing before or during cleaning *cannot* be used to determine whether a tank has been in leaded service.

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A lead-in-air analysis should be made *after* the tank has been cleaned, after all sludge and loose materials have been removed, and after the tank is essentially dry and free of liquids or puddles.

Flammable vapor analyzers do not give an indication of organic lead concentration and must not be used as a substitute for a lead-in-air analyzer.

D.4.3.1 Lead-in-Air Sampling Procedure and Location

The person making lead-in-air tests shall be a qualified person. The person should wear positive pressure full facepiece respirator (supplied air or SCBA) and protective clothing.

Lead-in-air tests should not be performed while forced ventilation is in progress, since test results may not reflect the real conditions that could be encountered during subsequent work. Forced ventilation should be stopped a minimum of 15 minutes before the test is performed.

To obtain representative samples of the atmosphere within the tank, the samples should be collected while the tester is walking near the inside periphery of the tank, but away from manholes, and in suspect areas such as near leaking tank bottoms. Areas in overhead or elevated positions such as buoyant sections of pontoons are not likely to pose exposure problems. The primary source of hazardous organic lead during cleaning is in the sludge at the bottom of the tank. There is no need to take lead-in-air samples outside the tank because the results will be well below permissible exposure limits.

D.4.3.2 Interpretation of Results

OSHA regulations at OSHA 29 CFR 1910.1000 Subpart Z, Toxic and Hazardous Substances 1910.1000, Table Z-1-A, limit personal exposure to the organic lead compounds tetraethyl and tetramethyl lead to an 8-hour time-weighted average of 0.075 milligrams per cubic meter (2 micrograms per cubic foot) and impose a skin designation.

D.4.4 SLUDGE DISPOSAL

EPA has identified "tank bottoms (leaded)" from the petroleum refining industry as a hazardous waste, No. K052, that requires special handling under the Resource Conservation and Recovery Act (RCRA). Implementing regulations are found in Title 40 CFR Parts 260 - 271.

Regulations enacted under RCRA require sludge from leaded gasoline tanks to be stored, shipped, treated, and disposed of in an environmentally approved manner. The general requirements of RCRA are noted here, but the regulations should be reviewed for compliance. Further, state requirements may be more stringent than EPA regulations and should be checked.

D.4.4.1 Notification and Application

All companies that generate, store, transport, treat, or dispose of hazardous wastes must file a form with EPA or the state agency implementing the RCRA program notifying the agency of those activities. Unless notification has been given to EPA or the state, waste may not be stored, transported, treated, or disposed of. All companies that store, treat, or dispose of sludge must apply to EPA for a permit to continue to do so.

D.4.4.2 Development of a Plan

Each company that generates sludge containing organic lead must develop a plan for the storage, treatment, and disposal of its sludge. If storage, treatment, or disposal is to occur on-site, the generator is then considered, for all practical purposes, to be the operator of a hazardous waste management facility and must obtain a permit to operate such a facility. If the company decides to dispose of the sludge off-site, no permit is required if the company stores the material in appropriate containers and ships it off site within 90 days of isolation, accumulation, or separation; however, EPA Title 40 CFR 262.34 specifies storage requirements before shipment.

D.4.4.3 Shipping Hazardous Waste

Before shipping an EPA-designated hazardous waste to an off-site facility, the company that generated the waste is required to prepare and sign a manifest that identifies the company, identifies the waste by its EPA and Department of Transportation hazardous waste number and name, identifies the off-site facility that will handle the material, and specifies the total quantity in the shipment. The company should be certain that the transporter and the selected facility have EPA identification numbers and permits to engage in hazardous waste management activities.

D.4.4.4 Recordkeeping and Reporting

The regulations impose extensive recordkeeping and reporting requirements. Companies that generate sludge must maintain copies of all manifest documents and records and must also submit annual reports.

D.4.4.5 Requirements for Transporters

Persons engaged in the off-site transportation of the sludge must comply with EPA's (and/or the applicable state's) specific regulations for the transportation of hazardous wastes, which govern notification, manifest, and record keeping. Persons transporting the sludge must also comply with the Department of Transportation regulations set forth in Title 49 CFR Parts 171 - 180.

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D.4.4.6 State Programs

RCRA authorizes EPA to delegate the implementation of the hazardous waste program to states under state authorities in lieu of the federal RCRA program. All but four states are authorized to administer the hazardous waste program. In authorized states, state laws and regulations replace RCRA and its regulations. Any state whose program has been approved by EPA may itself carry out the functions delegated to EPA under the act and may specify additional and more rigorous requirements. Consequently, companies that generate sludge and plan to dispose of sludge should ensure that plans comply with state programs.

D.4.4.7 Specific Facility Standards

Specific requirements governing storage, treatment, transportation, and disposal of hazardous wastes are updated continually by EPA and the states, and operators should consult the most up-to-date publications for details.

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APPENDIX E—PLANNING CHECKLIST

This appendix provides a checklist that may be useful in developing a site safety plan. It is not intended to be a complete plan since requirements for each tank will vary according to company policy, procedures or programs, tank design, product stored, tank location, environmental concerns, and the reason for cleaning the tank. The owner/operator should develop a plan for the tank that is to be taken out of service, the work to be conducted, and the recommissioning of the tank.

Figure E-1 provides a diagram illustrating the elements of a storage tank work plan.

E.1 Preparation

- Prepare a safety and work plan for the entire job, from decommissioning, cleaning, maintenance work, to recommissioning.
- Establish the duration of the job, starting and completion dates.
- Establish the reason for cleaning, such as to remove contaminants, to change product, to inspect, to make modification, to repair, to dismantle, or to dispose of waste.
- Conduct a survey of the site to determine the scope of work. Be familiar with area hazards before visiting the site.
- Obtain and review engineering drawings, such as plot plans, tank and piping details, fire protection, electrical facilities, and access roads.
- Assess potential safety and health hazards for the job. This should include a review of product information (MSDS or equivalent information) stored in tank to be entered and in adjacent tanks. Identify hazards unique to the job, for example, leaking tank bottom, faulty tank roof, minimum fire protection at the site, congested area, or vehicular traffic concerns.
- Confirm area electrical classification for equipment needed on the job, such as, Class I, Division 1, Group D or greater (A, B, or C).
- Determine what equipment will be needed, such as ventilating equipment, temporary pumps, vacuum trucks, and artificial lights, if any. Determine location of temporary equipment.
- Establish the need for respiratory and personal protective equipment.
- Determine the availability and location of shower and change room facilities.
- Determine how remaining product will be removed from the tank.
- Decide how sludge will be removed and how and where it will be disposed of. Check with appropriate environmental authorities for proper disposal.
- Determine who will perform the work. If a contractor or contractors will be used for some or all phases of the work, select a qualified contractor. The owner/operator tank administrative controls and safety rules should be a part of the contract.
- Establish who will be responsible for the safety of workers at the site: the owner/operator organization, the contractor organization, or both.
- Establish lines of communication at the job site between contractor and owner/operator.
- Review the work permit system, type of permits needed, signatures required, and how permits will be executed.
- Establish an emergency response system: medical aid, fire, and rescue.
- While the tank is still in service, lower legs (if floating roof) preferably when tank is full. Consider the integrity of the roof. Is the roof safe to walk on? Atmospheric tests must be performed before workers go onto the top of floating roofs.
- Pull and check for integrity a representative number of the roof legs.

E.2 Emptying the Tank

- For a floating roof tank, verify the integrity of the roof to avoid damage to the roof when being emptied.

Note: An important consideration when emptying a tank with an external floating roof is that the roof should be landed carefully on its legs. Also, roof drain hoses can become kinked or blocked. Blocked drains can cause water to accumulate on the roof, possibly causing roof damage or the roof legs to puncture the tank bottom. Therefore, it is very important to check roof drains when the roof has been landed.

- Use existing lines to empty the tank until the main pump nearly loses suction.
- If a temporary field pump is used to complete the emptying process, check the integrity of pump and hose and examine the safety of the system (for example, check valves to prevent backflow).
- Ensure that the field pump motor and all electrical connections meet area electrical classification.

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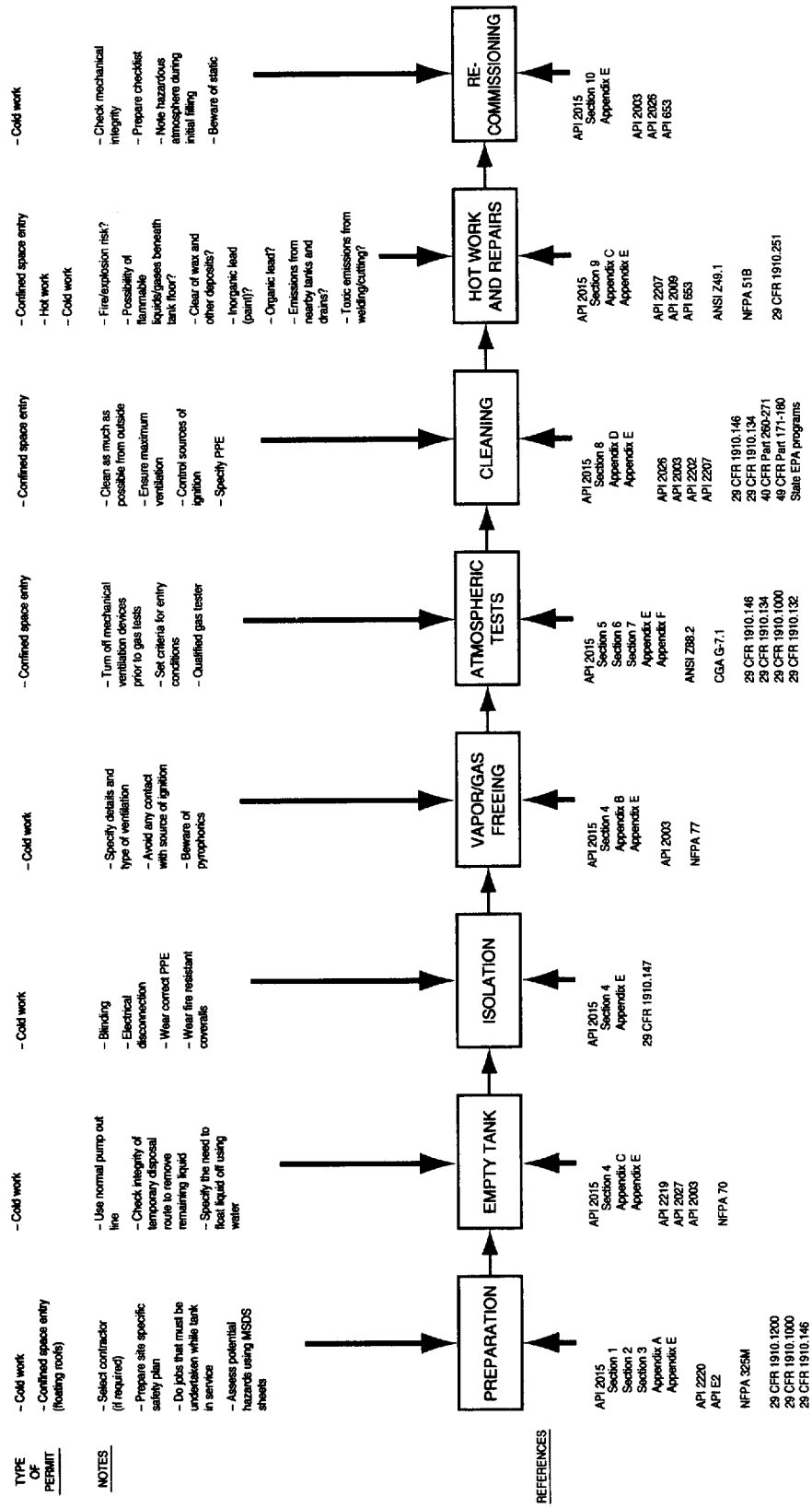


Figure E-1—Storage Tank Plan Elements

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- If atmospheric test results for the diked area are acceptable and the permit system is satisfied, move and position the temporary field pump and hose into place.

- Hook-up the field pump hose to an appropriate tank connection, such as a water draw line.

- Confirm that temporary equipment is electrically bonded and grounded.

- Pump out remaining liquid with the field pump until it loses suction.

- As an alternative method for removing remaining product, consider pumping water into the tank. Remaining product will float on water and can be removed through tank discharge line.

CAUTION: To avoid static, do not splash fill.

- If the remaining liquid is removed by vacuum truck, consider that vacuum trucks can create hazards: the engine is an ignition source; the vacuum process can create static electricity; and vapors discharged from the vacuum truck can pose problems. The location of the truck (upwind of tank) is important.

E.3 Isolating the Tank

- Prepare documents required under the permit program and the lockout and/or tagout procedure. This should include the isolating device and blind listing.

- Close all suction and discharge lines at the tank and install lockout/tagout devices.

- Lockout or tagout all energy sources, such as pump motors, mixers, and instrumentation devices.

- Isolate and/or disconnect the steam heating coils.

- Control ignition sources before opening any lines.

- Review thermal relief requirements for piping being isolated, to guard against overpressuring piping and related equipment.

- Select and use appropriate personal protective equipment during opening and blinding of lines.

- Proceed with isolation according to the isolating plan and blind list plan. Isolation may include tagging if specified on the work permit.

E.4 Vapor-Freeing the Tank

There are a number of ways of vapor-freeing a tank. In the case of large tanks it is accepted practice to ventilate using air movers, even though for many light products this process will entail passing of the tank atmosphere through the flammable range. Steam or inert gas ventilation methods should be avoided when suitable alternatives are possible.

- The ventilation process can be critical. Control ignition sources. Do not ventilate during thunder and lightning storms.

- Use air-driven mechanical ventilation if possible. Park the air compressor in a safe location.

- Ensure that ventilating equipment meets NFPA 70, Class I, Division 1, Group D (or greater) requirements and that equipment is electrically bonded to the tank.

CAUTION: If the tank has contained "sour" products (sulfur contaminants) be alert for pyrophorics which may ignite. A water spray system may be required to keep surfaces wet.

- Verify that energy sources are locked and/or tagged out (pump motors, mixers, gauging equipment) and that lines are blinded and/or isolated.

- Shut down all hot work before opening the tank.

- Verify that the tank is ready for opening. Execute work permits.

- Use appropriate personal protection for opening the tank. Take special precaution to prevent vapors from escaping when opening a side manhole.

- Have fire suppression equipment, including fire extinguisher and/or fire hoses ready.

- Consider installing ventilating equipment so vapors can discharge upwards and not settle on the ground. Start ventilating equipment at low capacity (so tank shell or roof does not collapse) to create a slight negative pressure in the tank. This procedure will prevent vapors from escaping at ground level. As manholes are opened, increase ventilating flow to accelerate the process.

- If installed in side manway, take precautions to prevent vapors escaping during installation. Consider discharging vapors upwards (via ducts, at least 12 feet above ground level).

E.5 Atmospheric Testing

- Verify that the person assigned to conducting atmospheric testing is qualified.

- Determine the frequency and type of tests, including oxygen content, flammable vapors and toxic substances, such

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as hydrogen sulfide, benzene, organic lead (organic lead tests must be done after the tank is clean, not before or during cleaning).

Note: Inorganic lead testers can not be used for organic lead testing.

- Verify that test equipment is appropriate for the atmosphere to be tested, is calibrated, and is properly checked.

- Select appropriate personal protective equipment for tester.

- To conduct tests, consider the following sequence: From outside the tank, test vapors being discharged from tank; if results are acceptable, shut down ventilation system at least 15 minutes to establish equilibrium inside the tank. Test through the tank shell manhole. If results and conditions are acceptable, issue an entry permit to allow tester to enter tank. The permit will specify personal protective requirements such as breathing equipment, clothing, gloves, boots, life line, and attendant, etc.

- Test the tank interior, but away from manholes. Test the atmosphere, especially in low areas and in critical and suspect areas, such as leaking tank bottom, roof supports, sumps, roof seals, roof drains.

- For floating roof tanks, test above the floating roof and inside pontoons. This procedure can be done by descending onto roof and/or by inserting a sensing wand through openings in roof.

- Be constantly aware that hydrocarbon vapors are heavier than air and tend to settle in low places.

Note: An inert atmosphere will produce false readings unless special gas testers and procedures are used.

- If test results and tank conditions are acceptable, issue the entry permit for work to begin. If test results are not acceptable, restrict entry, continue ventilation, and repeat testing steps.

E.6 Cleaning the Tank

- Arrange for disposal of sludge.

- Select appropriate personal protective equipment. If fresh air respiratory equipment is used, ensure that workers do not remove masks while in tank because of discomfort.

- Execute entry permit if tests results and tank conditions are acceptable and conditions are safe for entry, ignition sources have been controlled, no lightning storms are threatening, and the integrity of the tank is satisfactory.

- Atmospheric testing should continue periodically, such as at the start of each work day, at shift change, and during the shift as conditions change due to temperature increase, agitation of sludge, leaking tank bottom exposed, piping leaks, hydrocarbon absorbed in roof seal, etc.

- If the tank has side manholes, consider the use of a water hose stream to start cleaning from outside the tank. The employer shall determine the personal protective equipment and vapor concentrations at which workers may safely begin cleaning the tank from outside the manway without entering the tank. Ensure that hose nozzles are electrically bonded to the tank. Such cleaning may require an entry permit if personnel must break the plan of the manway.

- Remove loose sludge.

- When necessary, enter the tank and wash and remove waste materials. A vacuum truck may assist in this.

- As necessary, scrape or hydro-blast those internals to be repaired.

- Tank nozzles, columns, floating roofs, and support members are capable of trapping hydrocarbons and should be cleaned.

- Be alert to a leaking tank bottom with entrapped product.

- Clean areas above the floating roof; check flotation compartments. Waste materials should be removed.

- Ensure continuous ventilation throughout the cleaning process.

- Maintain control over potential ignition sources.

- In warm climates, monitor workers for heat stress.

E.7 Hot Work

- Where hot work will be done, atmospheric tests must be taken in surrounding and immediate area of work.

- A hot work permit should be issued after acceptable atmospheric test results are obtained. The permit should specify safety and health requirements, such as personal protection and need for a fire watch, fire extinguisher, fire hoses, blankets, and so forth.

- For work inside the tank, survey areas for hazards such as entrapped hydrocarbons beneath leaking tank bottoms, in floating roofs, seals, pontoons or roof supports, etc. For work outside, cover sewers and drains, and, if applicable, remove or temporarily cover hydrocarbons absorbed in soil.

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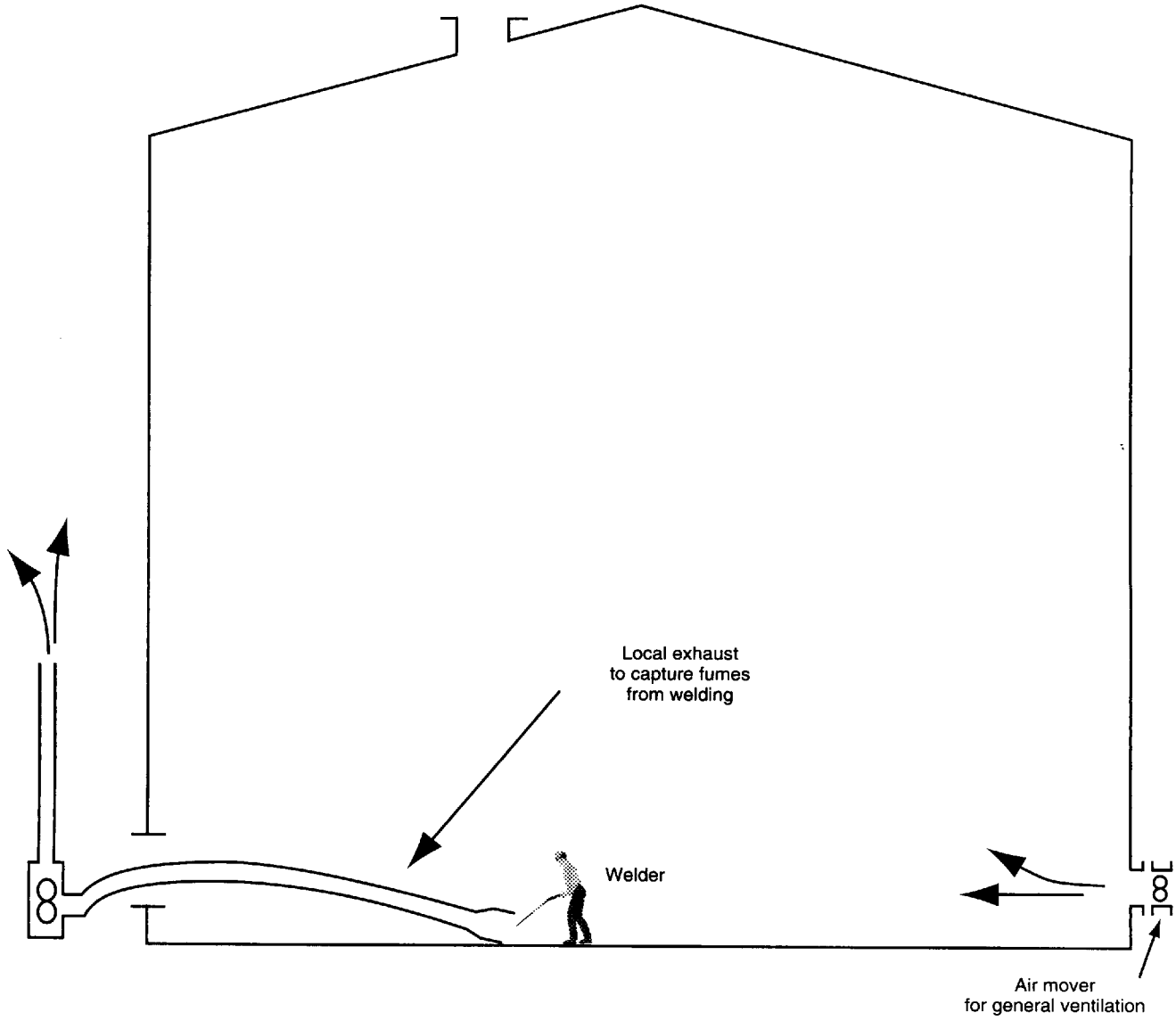


Figure E-2—Example of Ventilation for Welding Fumes During Hot Work

- For work on tank bottoms, refer to API Publication 2207.
- If work is inside the tank, consider the need for a communications system with a person stationed outside the tank.
- Consider the need for local or space mechanical ventilation while workers are inside tank. See Figure E-2.
- Provide protection as needed from heated materials, such as toxic vapors from welding rods or coatings, heating of metals, dusts from wire brushing or grinding; evaluate the need for eye, respiratory, face, skin or hearing protection.
- If hot work is done in tanks that contained leaded gasoline, first scrap to bare metal an area at least 12 inches (30 centimeters) on each side of the heated metal or wear supplied air respiratory equipment and appropriate protective clothing.
- Consider the need for exposure monitoring. For example: OSHA 29 CFR 1926.62 requires exposure monitoring un-

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less an initial assessment has been made under conditions closely resembling the processes, types of material, control methods, work practices and environmental conditions present.

- If hot work is done outside of a tank with lead based paint, wear appropriate respiratory protection and appropriate protective clothing.

- Stay within requirements of the permit system.

- Tanks that contain waxy substances and other heavy oils can emit flammable vapors and be ignited. Therefore, the area in which hot work will be performed must be thoroughly cleaned.

- Visual inspections must be undertaken for cleanliness of walls and bottoms, together with atmospheric tests.

E.8 Recommissioning the Tank

- Survey the tank entirely to ensure it has been cleaned, all repairs completed, floating roof components are in place, and all internals have been replaced. Look for tools, scraps, and rags.

- Close all manways.

- Revoke all unexpired permits.

- Check with the inspection department that any required mechanical integrity tests have been completed (for example, a hydrotest).

- Remove all blinds and tags used for lockout or tagout procedure.

- Verify that all tags and blinds are removed and that all flanges and manways are bolted properly.

- Sign-off that the lockout/tagout for this job is complete.

- Review any check or punchlists for all remaining open items after completing a recommissioning safety check of the following items:

  Inspection
  — The tank foundations have been inspected and confirmed to be in satisfactory condition.
  — The dike walls have been reinstated.
  — The dike drainage system is serviceable and clean.
  — Necessary repairs to the tank have been carried out.
  — Tank fittings and valves have been correctly reinstalled and are serviceable.

  Electrical
  — The tank shell is grounded/earthed.
  — Any ground/earth cables associated with the floating roof or internal floating cover are securely connected and intact.
  — Any seal grounding/earth (bonding) shunts on a floating roof tank are correctly installed and intact.
  — Any tank or associated piping cathodic protection system has been reinstated.
  — Electrical fittings associated with any side entry mixers have been correctly installed.

  Instruments
  — Tank level gauging equipment has been reinstalled and reconnected and is functioning correctly.
  — Any separate high or low level alarms or both have been reinstalled and are functioning correctly.
  — Any tank temperature measurement instrumentation has been correctly reinstalled.
  — Any instrumentation associated with the side entry mixers has been correctly replaced.

  Fire Protection
  — Any foam lines and their associated fittings have been correctly reinstalled.
  — Any sprinkler system is correctly reinstalled.
  — Systems have been live tested to meet requirements of pre-fire plan and codes/standards.

  Mechanical
  — Materials and debris from tank cleaning operation and subsequent inspection and any repair work have been removed from inside the tank, from the roof of the tank, from any internal floating cover, and from the vicinity of the tank for appropriate disposal outside the dike area.
  — Any internal floating cover has been installed in accordance with the manufacturer's requirements.
  — Any side entry mixers have been correctly reinstalled.
  — Handrailings on stairways and around the tank roof or wind girder are intact and secure.
  — The roof ladder is correctly positioned and intact.
  — The roof legs are all in position and correctly set.
  — The automatic vent legs are correctly installed and have been tested.
  — The roof drain sumps are all clear of debris, and all non-return/check valves are operational.
  — All pontoons are clean internally, and all pontoon manway covers have been replaced correctly.
  — Any free vents, at the top of the shell or on the tank roof, have the correct "free area" mesh/screen fitted and are clear and free from blockage.
  — All roof manway covers and other roof fitting covers are closed.

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Operational

— The tank has been inspected internally and is acceptable for service.
— The tank has been inspected externally, including the roof and any internal floating cover, and is acceptable for service.
— All roof drains, water draw-off, suction and discharge valves are serviceable and properly identified.
— All emergency roof drains on single roofs have been primed with water. These should be blinded off if shown to be unnecessary.
— All emergency roof drains on double deck roofs are fitted with the correct mesh screen and are clear.
— The tank drain valves are closed.
— The roof drain valve is placed in the open or closed position in accordance with local policy/procedures.
— The dike drain valve is tightly closed.

E.9 Initial Filling

- Minimize the number of personnel in the area immediately adjacent to the tank during the filling operation especially if the filling operation is for the purposes of hydrotest.
- Line-up tank for filling.
- Begin filling tank slowly until tank suction line is well submerged to avoid turbulence and generation of static. The open area above the liquid will go through the flammable range if low flash product is being pumped. Maintain control of ignition sources. Do not fill during a lightning storm. If tank has a floating roof, fill tank slowly for the same reason and to avoid canting of roof until roof lifts off its support legs. Personnel should not go onto the top of internal floating roof tanks or the floating roof of external floating roof tanks for a period of approximately 18 hours after refloating or until the vapor concentration above the roof is below 10% LFL and the atmosphere is safe to breath.
- Check tank and lines frequently for leaks during filling process, especially manholes and nozzles that were opened.
- Confirm that tank and dike area housekeeping is good.
- Complete required documentation.

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APPENDIX F—OXYGEN AND FLAMMABLE VAPOR ANALYZERS

This appendix provides additional guidance related to the use of oxygen and flammable vapor analyzers.

F.1 Oxygen Analyzers

For the purpose of this example calculation, the air external to the tank has been tested and contains approximately 20.8% oxygen. If a tank is contaminated with other vapors or gases, the presence of those vapors or gases will have a significant effect on the electrochemical oxygen sensor reading as follows:

$$20.8\%/100\% = X\%/100\%/(100-Y\%)$$

Where:

X = % by volume reading of the instrument for oxygen.

Y = % by volume of the contaminant gas present.

Examples:

If $Y = 0\%$ then $X = 20.8\%$.

If $X = 19.5\%$ then $Y = 6.25\%$. For most hydrocarbon gases, a reading of 6.25% would put the atmosphere in the flammable range.

If $Y = 1\%$ then $X = 20.6\%$.

If $Y = 10\%$ then $X = 18.7\%$.

If a gas had an LFL of 1.4% then 10%LFL = 0.14%.

If $Y = 0.14\%$ then $X = 20.77\%$ (20.8%).

F.2 Flammable Vapor Analyzers

The results of tests for flammable vapor will vary as the concentration/composition of the tank atmosphere varies from the calibration gases used to calibrate the instrument. Even though the instrument responds to any combustible gas/air mixture below the flammable range, that instrument provides accurate measurement only for the specific gas for which it has been calibrated. For other flammable vapors, the responses may be more or less sensitive than the calibration gas which is used. The user should consult the instrument manufacturer's literature for conversion factors to be used when testing for other flammable vapors.

Analyzers may not measure the low-level concentrations of hydrocarbons (in parts per million) that may pose potential health hazards. An analyzer may indicate a reading of zero when the concentration of toxic vapor is above the permissible exposure limit (PEL) or threshold limit value (TLV). An inert atmosphere or one with low oxygen content may cause flammable vapor analyzers to give false readings. A special gas-testing instrument and procedure is required to ensure valid results in inert atmospheres or if the oxygen content of the atmosphere being tested is not similar to ambient air.

Flammable vapor tests should not be performed during steaming operations because the results may be erroneous. A flammable vapor test should be done after steaming operations are complete and after the steam has condensed and atmospheric conditions have reached equilibrium.

The following factors may affect the accuracy of flammable vapor analyzers:

a. Instrument error. An instrument calibrated for pentane may give readings that are 70-80% too high in the case of hydrogen and carbon monoxide or 50% too low in the case of xylene.

b. Temperature variation. Atmospheres tested at elevated temperatures may give results which require additional factors of safety.

c. Poisoning. Certain contaminants may poison the instrument, resulting in no reading or low readings.

d. Gas or vapors above the upper flammable limit. When measuring atmospheres above the upper flammable limit, the instrument will move quickly to the flammable/explosive range and then the needle/reading will return to zero.

e. Presence of inert gas. Instruments have indicated an atmosphere free from flammable gas after nitrogen or some other inert gas had been used to purge out a flammable gas or vapor. Without the admission of air, the instrument may give false readings. As a result, it is important to take the oxygen reading first to ensure that oxygen is present.

f. Leakage of air into the instrument. Many instruments have a sampling line which is attached to the instrument body by a screw thread coupling. Unless this coupling is fitted properly and kept tight, air can be drawn into the sample gas, thereby diluting the sample gas and resulting in a false reading.

g. Use in oxygen-enriched atmospheres. The electrical protection certification of the instrument may not be valid for oxygen-enriched atmospheres.

h. Water vapor. Heated atmospheres which contain very large proportions of water vapor may result in false readings similar to that obtained from the effect of inert gas. There must be sufficient air present to give a correct safe reading.

i. High flashpoint materials. Flammable vapor analyzers will not detect flammable atmospheres at ambient temperatures of lubricating oil, diesel oil or fuel oil. However, flammable atmospheres may be generated during hot work if these materials are present and heated.

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